



PSYCH 201B

Statistical Intuitions for Social Scientists

Linear Mixed Models

*You can download these slides:
course website > Week 10 > Overview*

03/10/2025

Welcome to our LAST week together! 🥲 🎉

- A few reminders...
 - Proposal deadline: **Wed March 12th**
 - HW4 (available @midnight) deadline: **Mon March 17th**
 - Final Project deadline: **Thurs Mar 20th**
 - Submission details (also on wk10 overview page):
<https://tinyurl.com/final-instructions>

Today: Linear Mixed Models

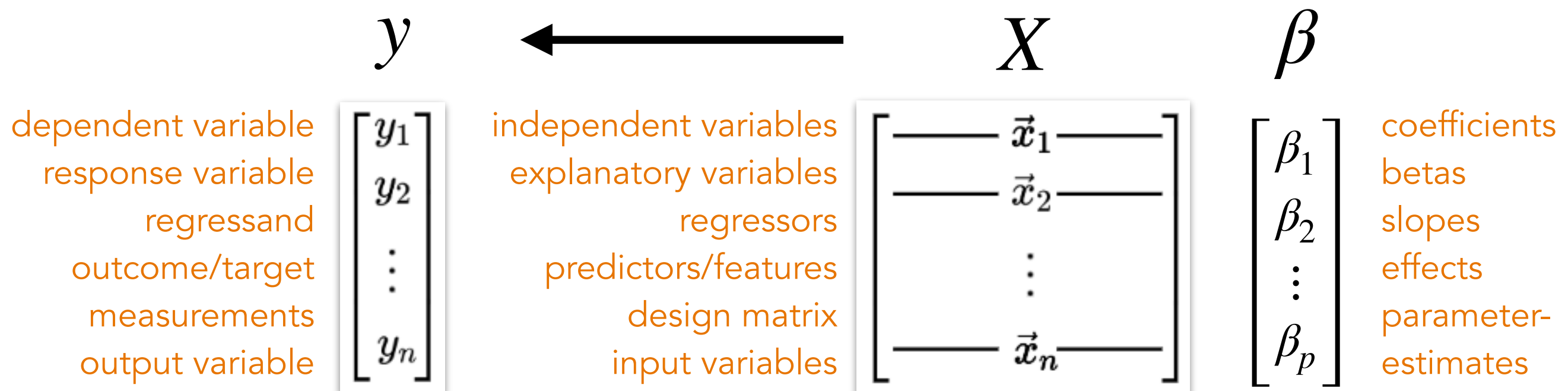
- Part 1: Slides Together
 - Violating independence in GLM
 - The solution: LMMs
 - LMM Essentials
- Part 2: Self-guided tutorial notebook

Review: The General Linear Model (GLM)

$$y = X\beta + \epsilon$$

Models a scalar variable y as a linear function of predictor variables X by minimizing:

sum of squared errors $\longrightarrow \epsilon^2 = \sum (y - X\hat{b})^2$



Review: The General Linear Model (GLM)

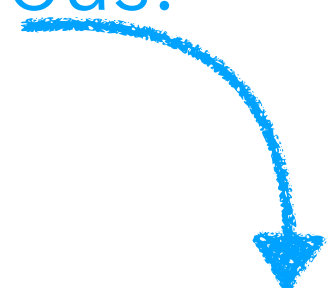
$$y = X\beta + \epsilon$$

All Categorical:

2 levels = t-test

3+levels = ANOVA

Categorical & Continuous:
ANCOVA



y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

independent variables
explanatory variables
regressors
predictors/features
design matrix
input variables

X

$$\begin{bmatrix} \text{---} \vec{x}_1 \text{---} \\ \text{---} \vec{x}_2 \text{---} \\ \vdots \\ \text{---} \vec{x}_n \text{---} \end{bmatrix}$$

β

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

coefficients
betas
slopes
effects
parameter-
estimates

Review: GLM Assumptions

Errors should be independent & identically **distributed (i.i.d)**

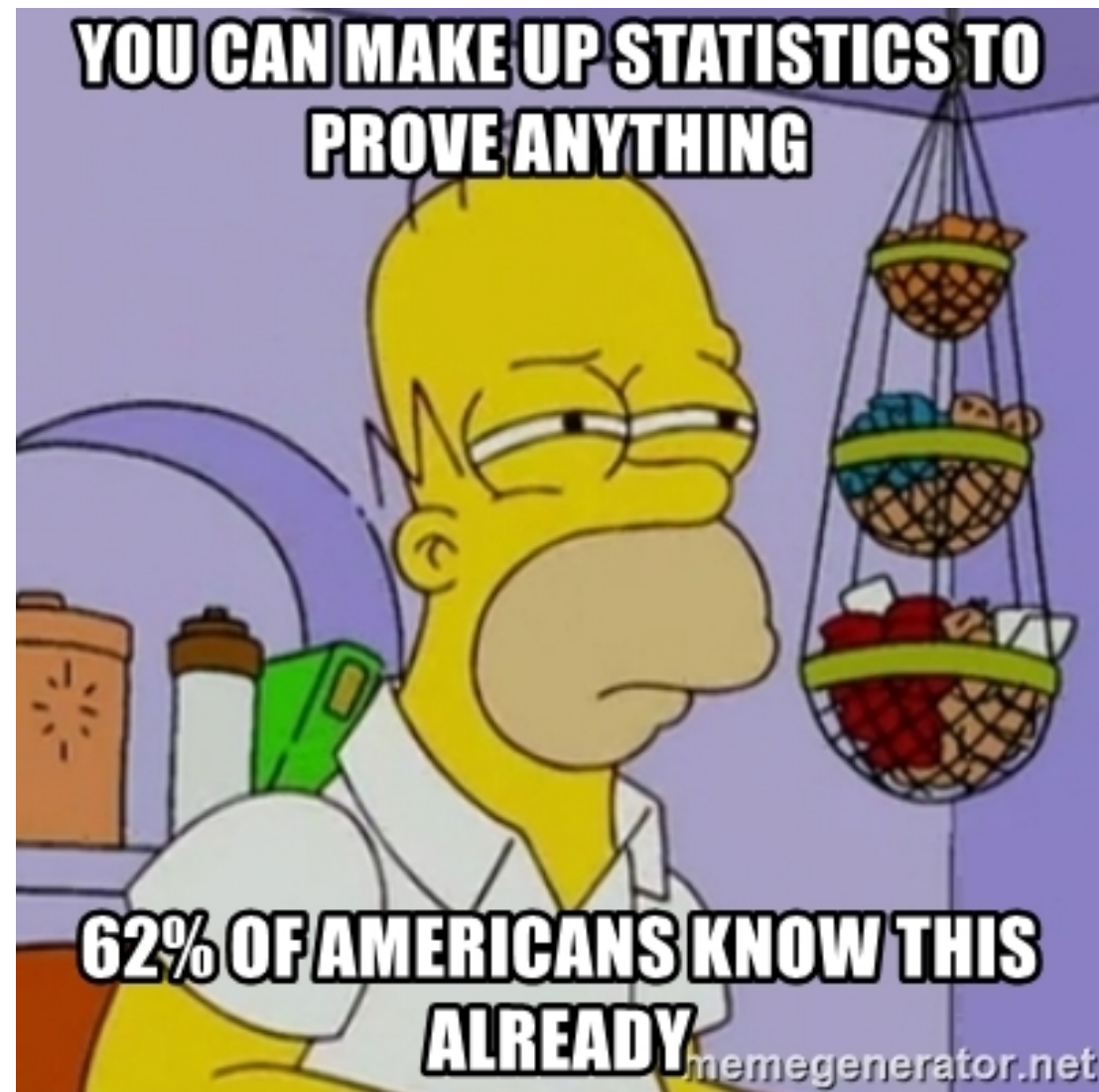
- **Normality:** assumes that the *residuals* are normally distributed. It's actually okay if the predictors and the outcome are non-normal, so long as the residuals are normal.
- **Additivity & Linearity:** assumes that the outcome **y** is a linear function of separate predictors **X** that are added together
- **Homoscedasticity:** assumes that the variance of our residuals doesn't change as function of our predictors; model errors shouldn't depend upon value of predictor
- **No perfect multi-collinearity:** assumes that our predictors **X** are not just linear combinations of each other; can't figure out what the *unique variance* each one explains
- **Independence of errors:** assumes our residuals don't depend upon each other - this really only gets violated when you have repeated-measures, time-series, geospatial, or multi-level

Violating Independence

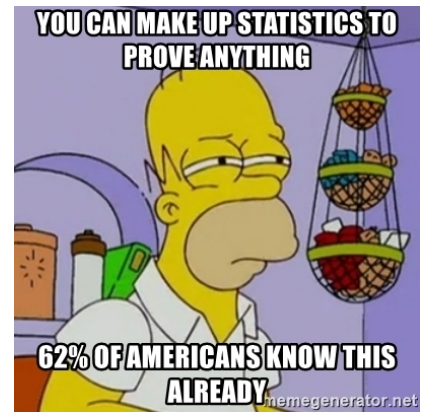
- So far, all the GLM flavors we've discussed (one or more continuous/categorical predictors) satisfy the independence assumption
- But you'll often work with data where it's violated, e.g:
 - **psychology experiments**: repeated measurements from the same individual
 - **survey data**: multiple measurements from the same city/county/state
 - **time series**: observation at time-point t depends upon $t - 1$

What's the big deal?
What could go wrong?

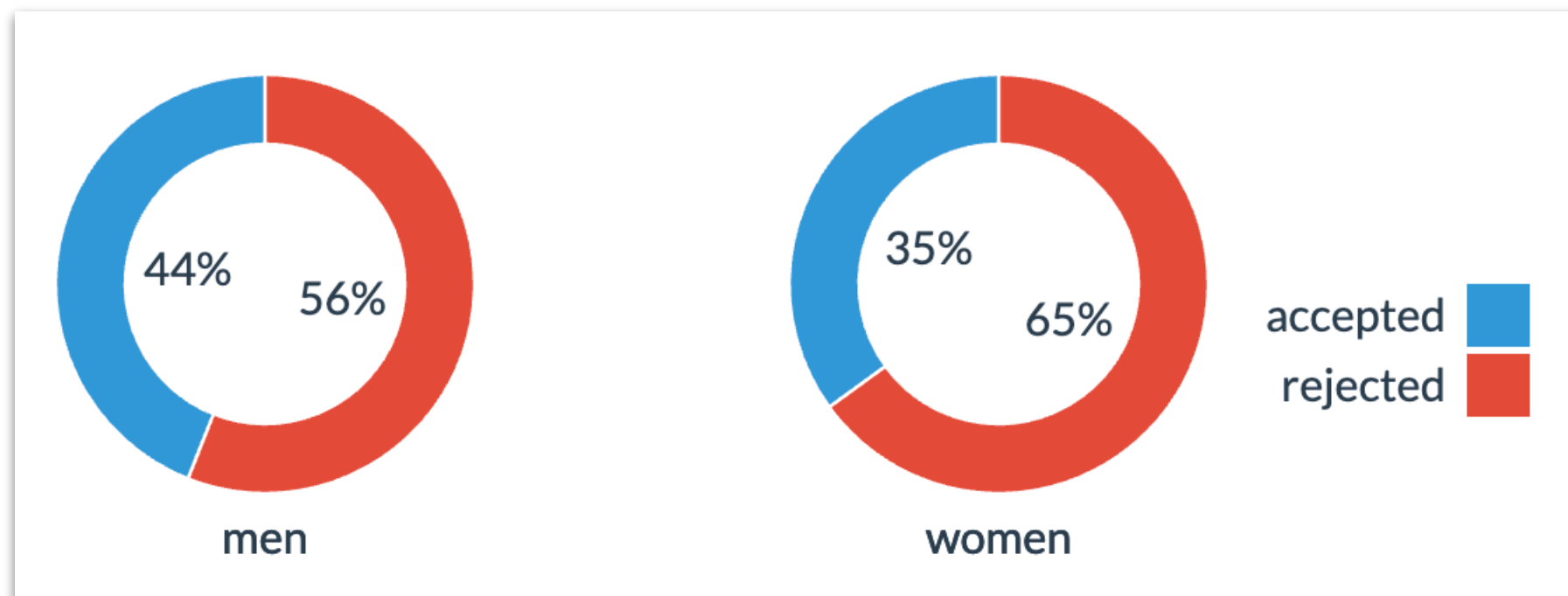
Simpson's paradox



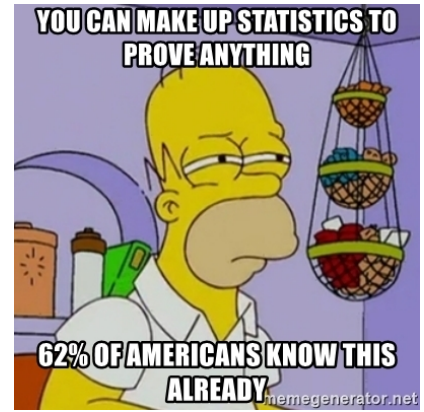
Simpson's paradox



In 1973, the University of California-Berkeley was almost sued for sex discrimination. The numbers looked pretty incriminating: the graduate schools had just accepted 44% of male applicants but only 35% of female applicants.



Simpson's paradox

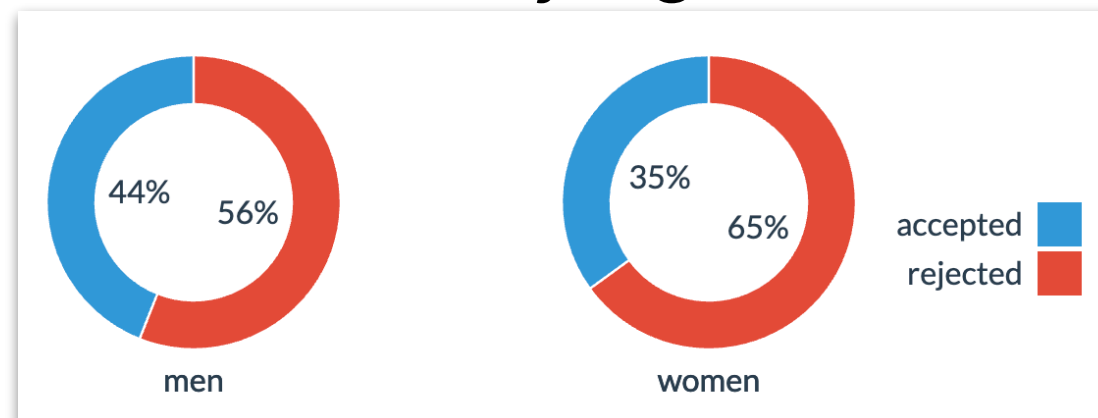


In 1973, the University of California-Berkeley was almost sued for sex discrimination. The numbers looked pretty incriminating: the graduate schools had just accepted 44% of male applicants but only 35% of female applicants.

When researchers looked at the evidence, though, they uncovered something surprising:

What?

*If the data are **properly pooled** by department... there is a small but statistically significant **bias in favor of women***



Simpson's paradox

Aggregated by Sex only

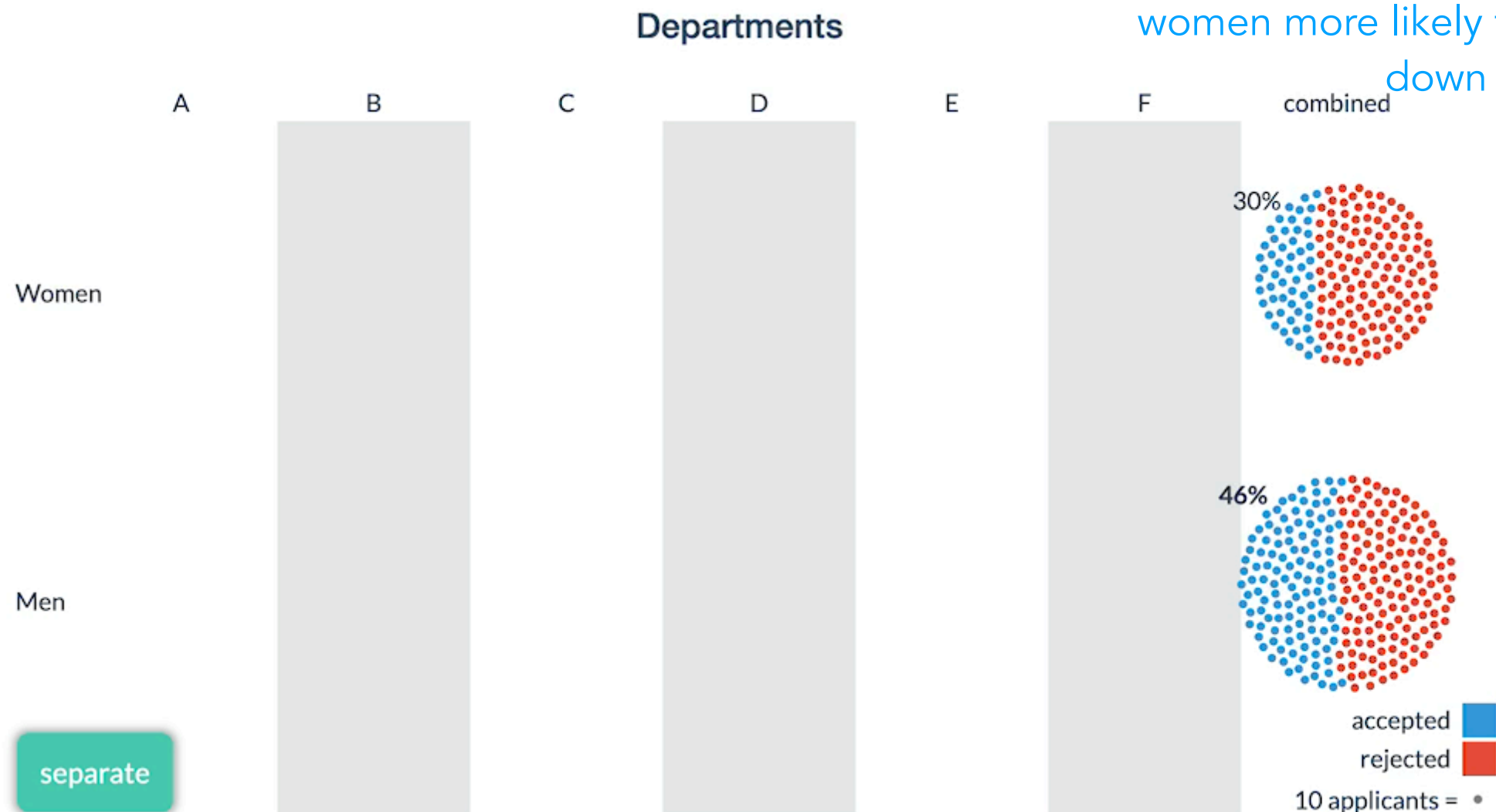
	Men		Women	
	Applicants	Admitted	Applicants	Admitted
Total	8442	44%	4321	35%

overall men are more likely to be admitted

Aggregated by Sex & Dept

Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%

women more likely to be admitted when broken down by department



Simpson's paradox

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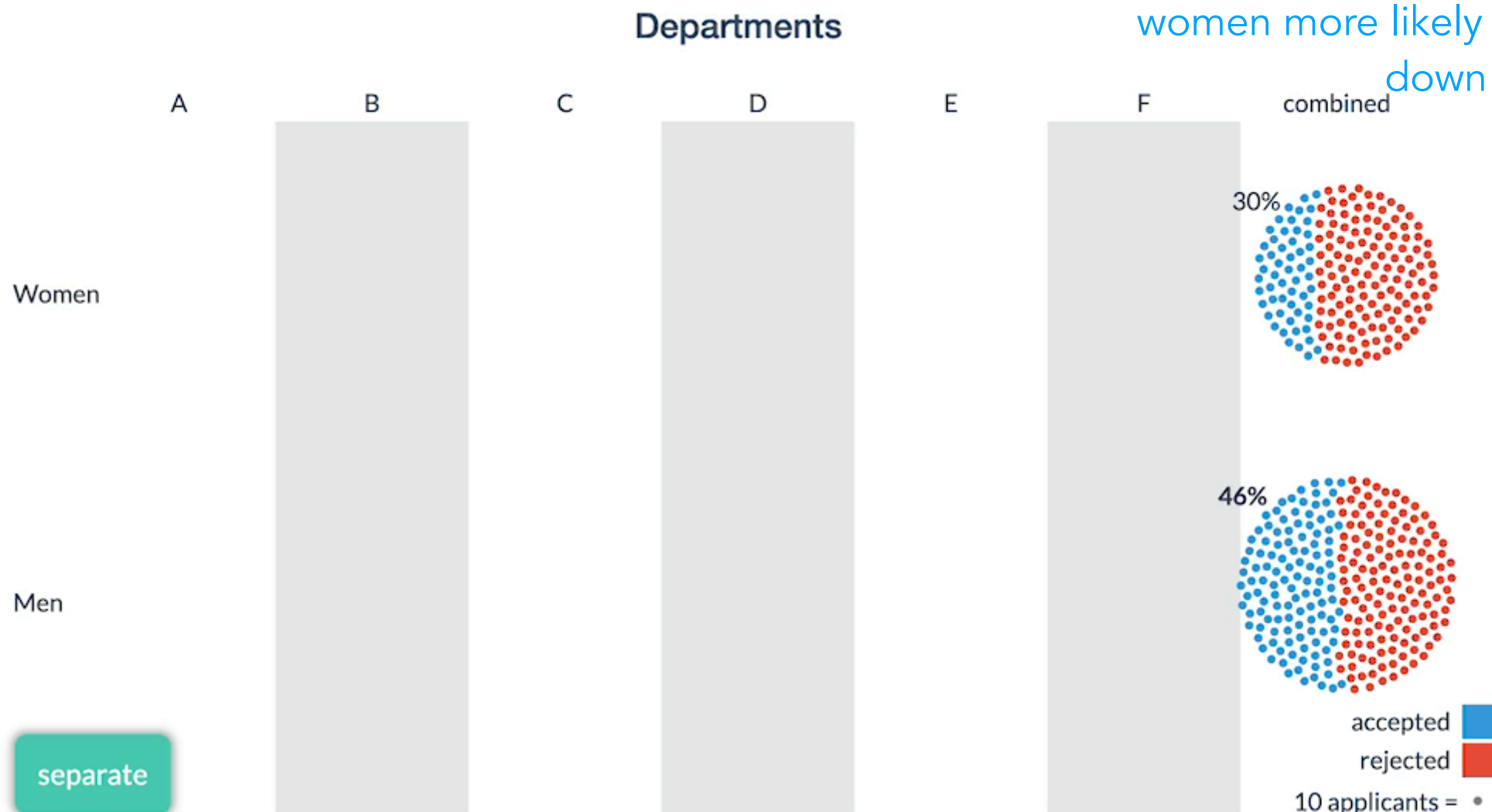
overall men are more likely to be admitted

↑ Ignores variance between departments!

Aggregated by Sex & Dept

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↑ Women applied to more competitive departments!

Simpson's paradox

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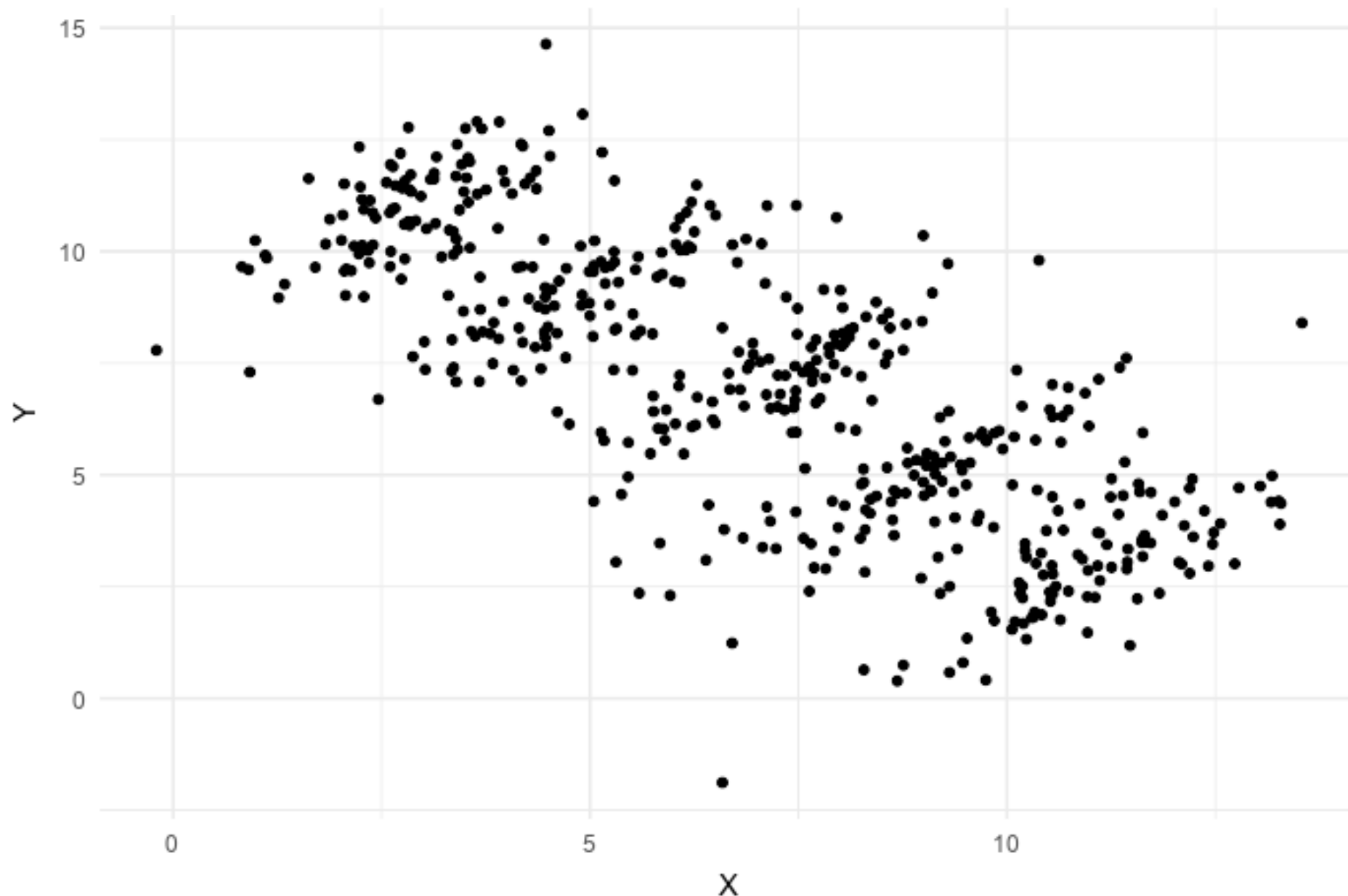
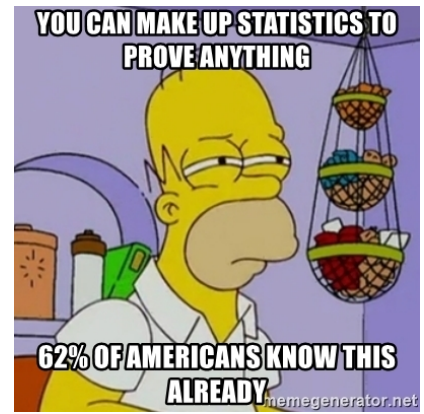
In fact, the pooled and corrected data showed a "small but statistically significant bias in favor of women."



Bickel, P. J., Hammel, E. A., & O'Connell, J. W. (1975). Sex Bias in Graduate Admissions: Data from Berkeley. *Science*, 187(4175), 398-404.

Simpson's paradox

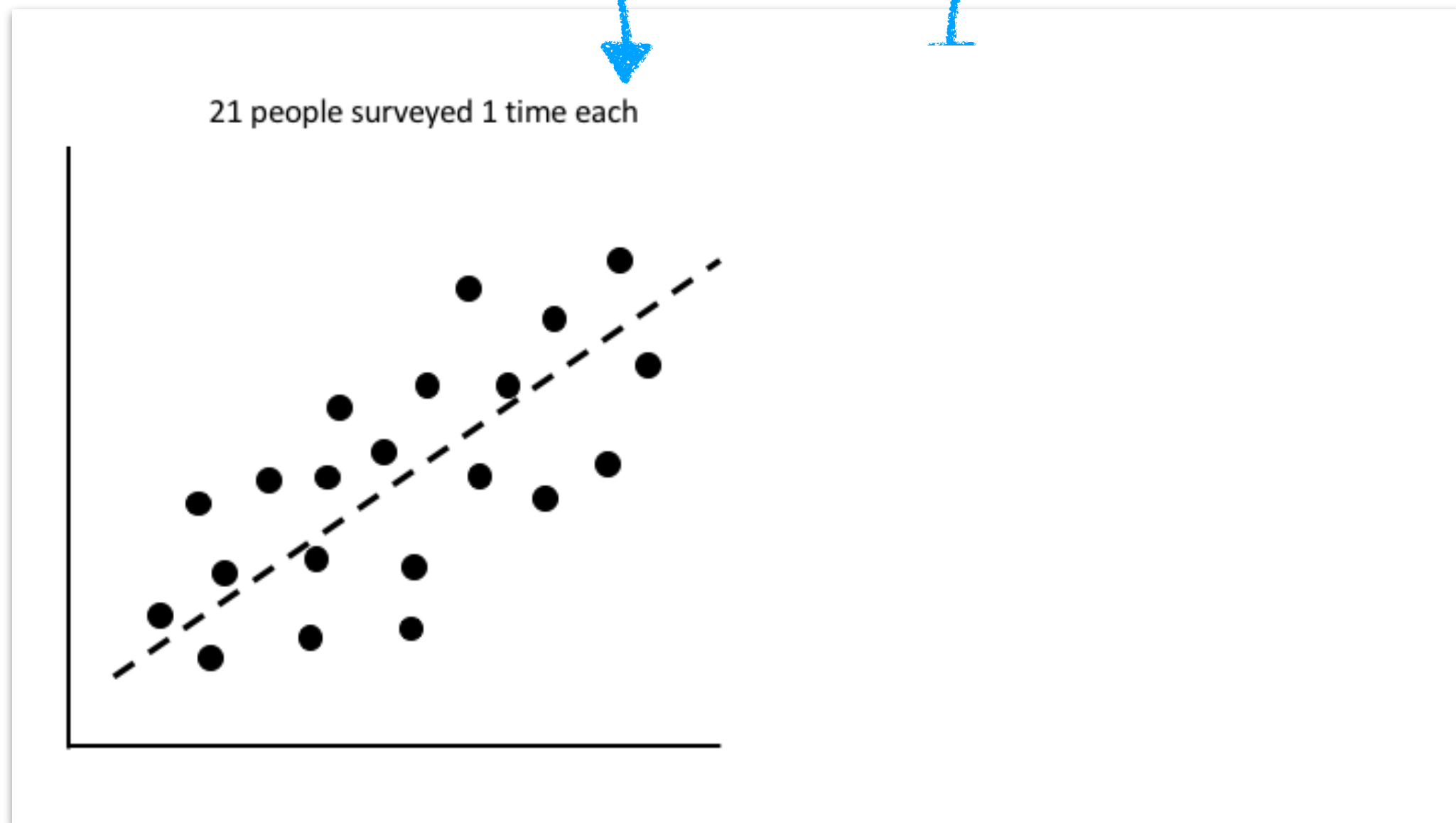
Phenomenon in statistics when a trend that appears in several **groups** of data disappears or reverses when the groups are **combined**



Dependent Data

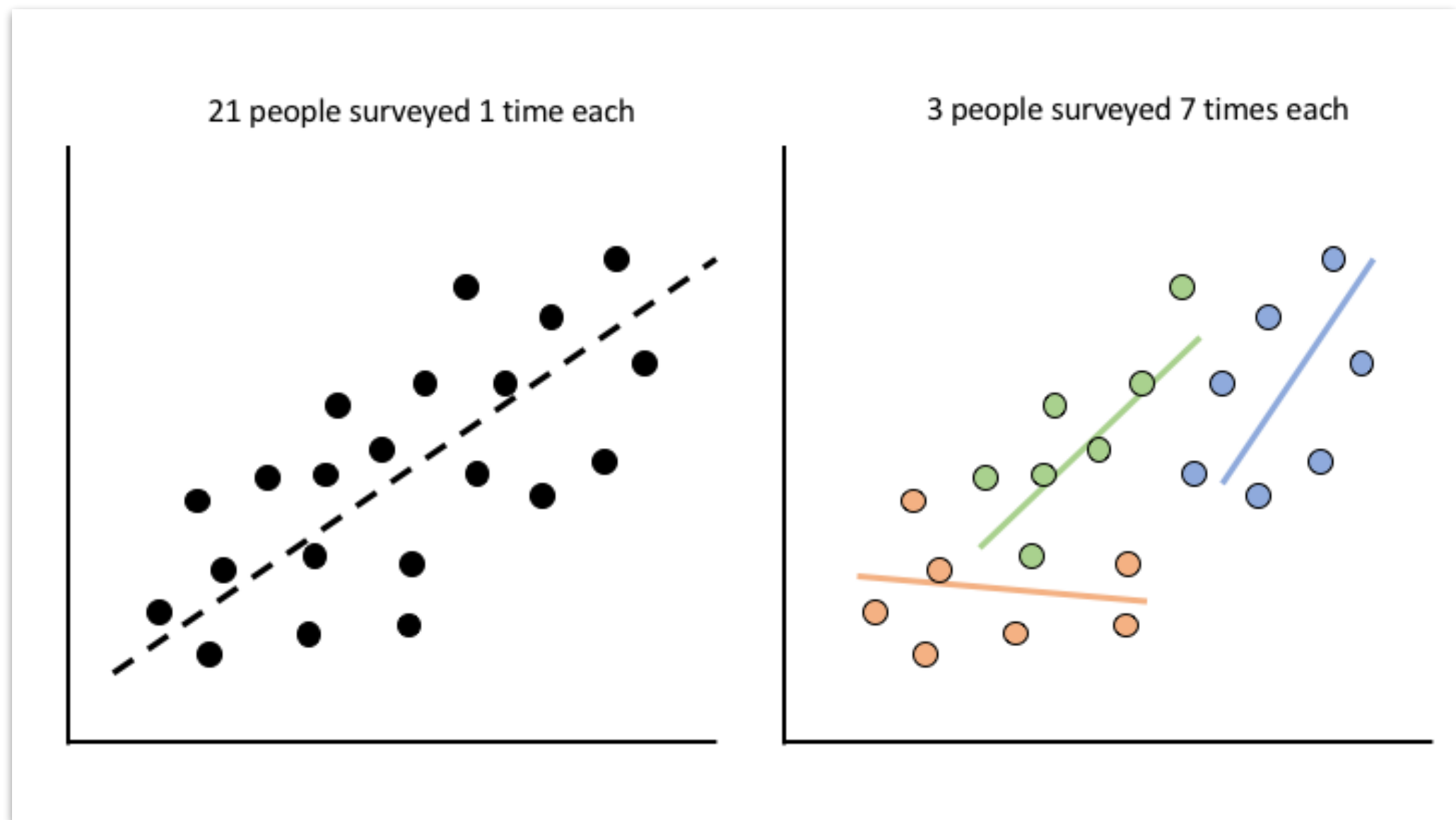
Each person's response is independent of everyone else's response

Data points *within* each person are more similar than data points across people



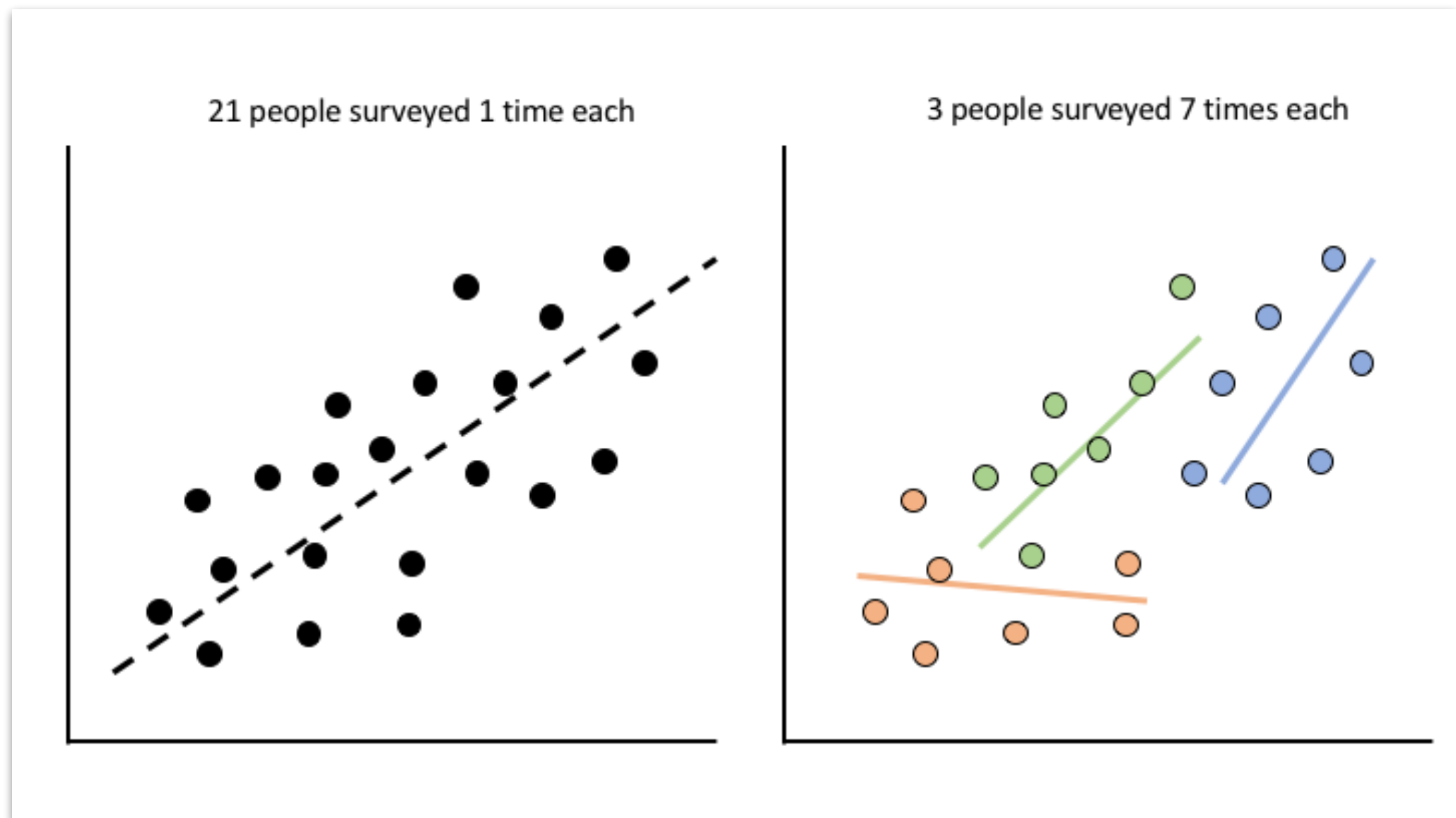
Perils of Ignoring Dependence

- Parameter estimates could be inaccurate or even reversed
- Standard-errors too small = t-stats too big = p-values too small = **our inferences are too liberal!**
- Higher Type I error (remember which one that is?)



Perils of Ignoring Dependence

- Parameter estimates could be inaccurate or even reversed
- Standard-errors too small = t-stats too big = p-values too small = **our inferences are too liberal!**
- Higher **False Positive Rate**

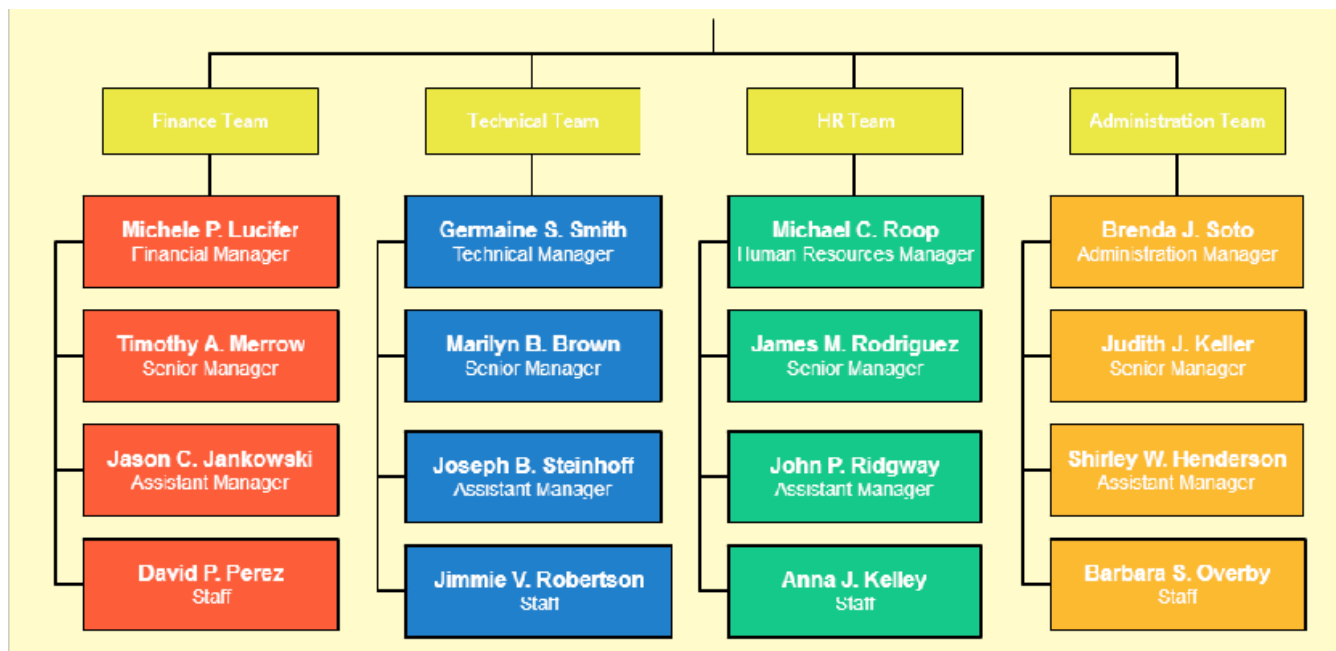


The solution: **Linear Mixed Models**

An extension of GLM that allows us to account for dependencies at data by estimating parameters at different *levels*

Hierarchical data

e.g. person *within* team *within* company

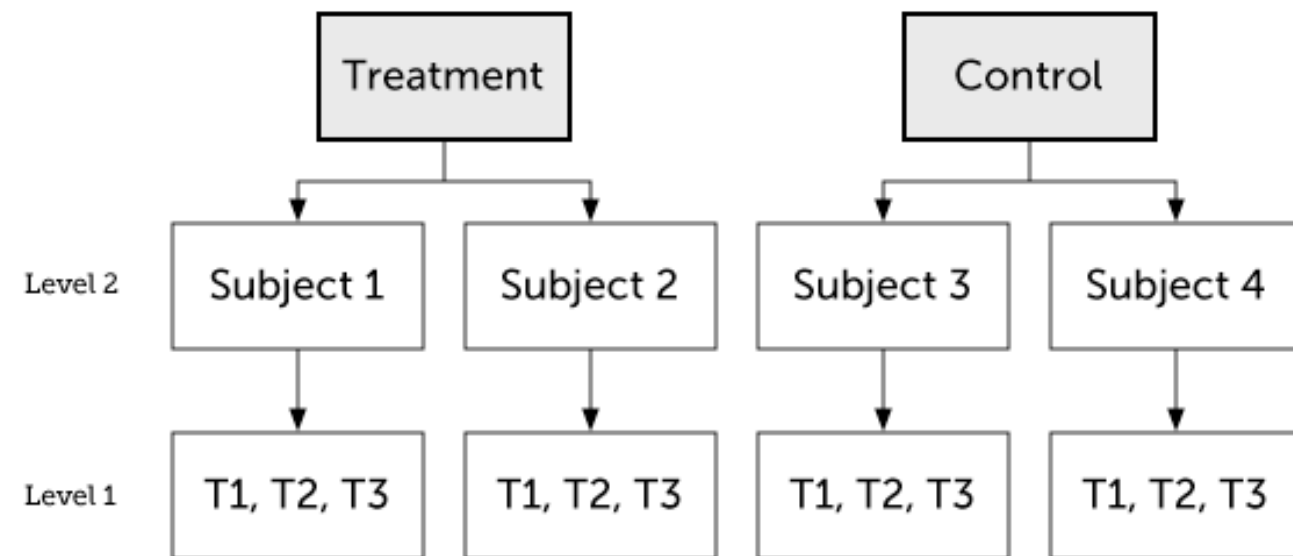


department > applicant

state > county > city

Longitudinal data

e.g. repeated measures per person



experimental condition > subject > trial

group > subject > time-point

The solution: **Linear Mixed Models**

An extension of GLM that allows us to account for dependencies at data by estimating parameters at different *levels*

Wait what about repeated-measures ANOVA?

Stop using them

- Anything you can do with rm-ANOVA you can do with LMM!
- With rm-ANOVA we *know* our model doesn't handle dependence explicitly so we have to *correct* for this violation!
 - "sphericity violation" Greenhouse-Geisser correction
- LMMs let us **explicitly model dependence in different ways**

The solution: **Linear Mixed Models**

It is a shame that psychology students don't really learn much about analyzing multilevel data. Think about studies you have read recently in psychology or neuroscience. How many of them take a single measurement on the DV from each participant? Very few, if any. Nearly all take multiple measurements, for one or more of the following reasons:

- (1) the researchers are measuring the same participants across levels of a factor in a within-subject design;*
- (2) they are interested in assessing change over time;*
- (3) they are measuring responses to multiple stimuli.*

*Multilevel data is so common that **multilevel analysis should be taught as the default approach in psychology.***

*Learning about multilevel analysis can be challenging, but you already know much of what you need by having learned about regression...**you will see that it is just an extension of simple regression.***



Dale J Barr
(big deal in LMMs & language)

Linear mixed effects models

Linear mixed effects models*

**Often taught as a full quarter class by itself!*

We'll try to cover the essential intuitions in 2 days 😊💧

Terminology & Jargon

- Linear Mixed Models are also called...
 - **H**ierarchical **L**inear **M**odel (HLMs)
 - **M**ulti-**L**evel **M**odel (MLMs)
 - Mixed Effects model
 - Growth Curve model
 - Random Effects model
 - Variance Components model

Different names for the same approach!

LMMs: The Essentials

1. Ways to aggregate dependent data
 - Complete, No-pooling, **Partial-pooling**
2. Parts of the mixed model
 - Fixed vs Random Effects
 - What's happening under-the-hood
3. Specifying Random Effects
 - Random Intercepts & Slopes
 - How to choose your random-effects
4. How to estimate LMMs
 - Meet Pymer4

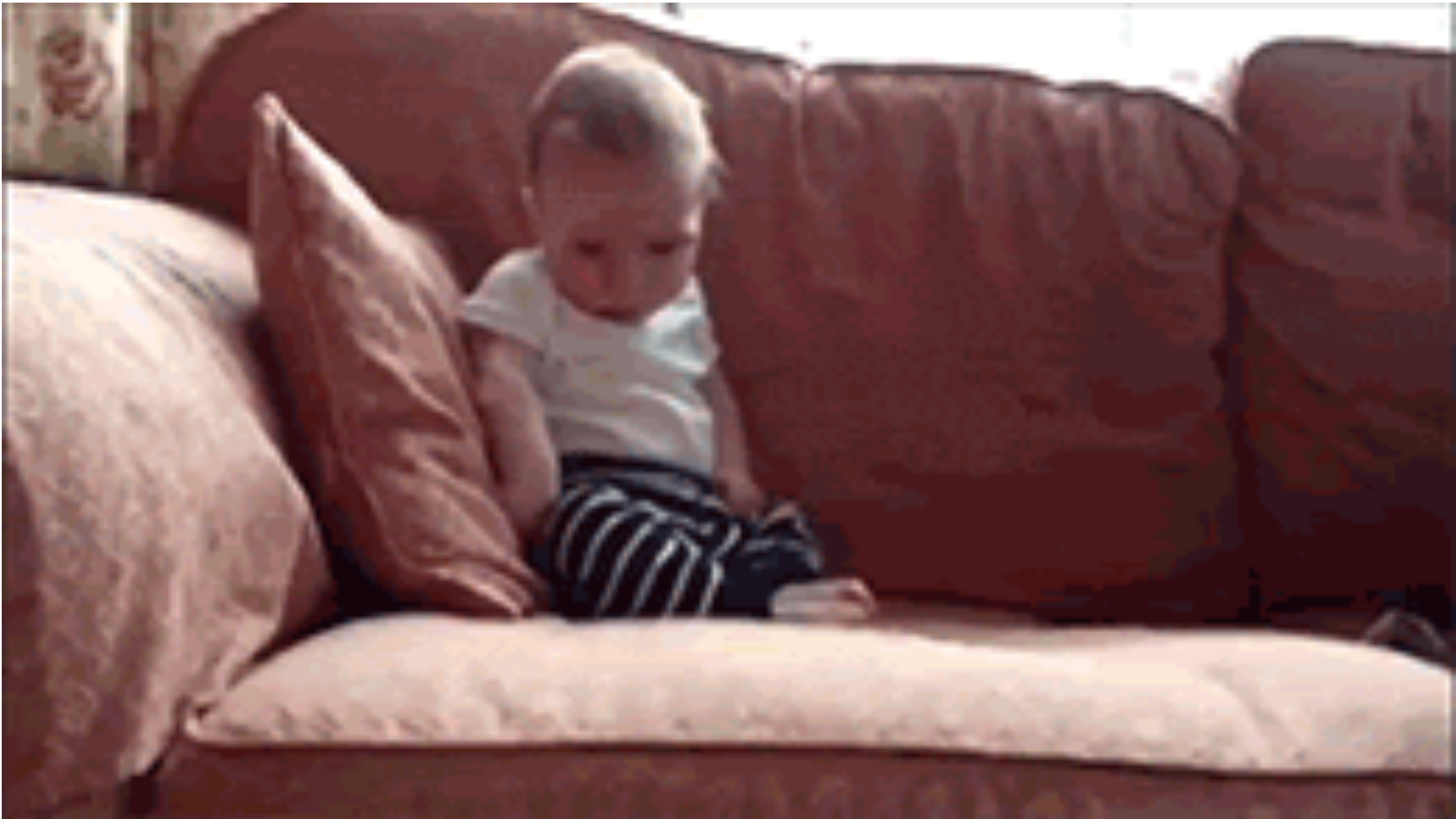
Let's build intuitions with a real dataset...

1. Ways to aggregate dependent data
 - Complete, No-pooling, **Partial-pooling**
2. Parts of the mixed model
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Data set

How does sleep deprivation affect reaction time?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
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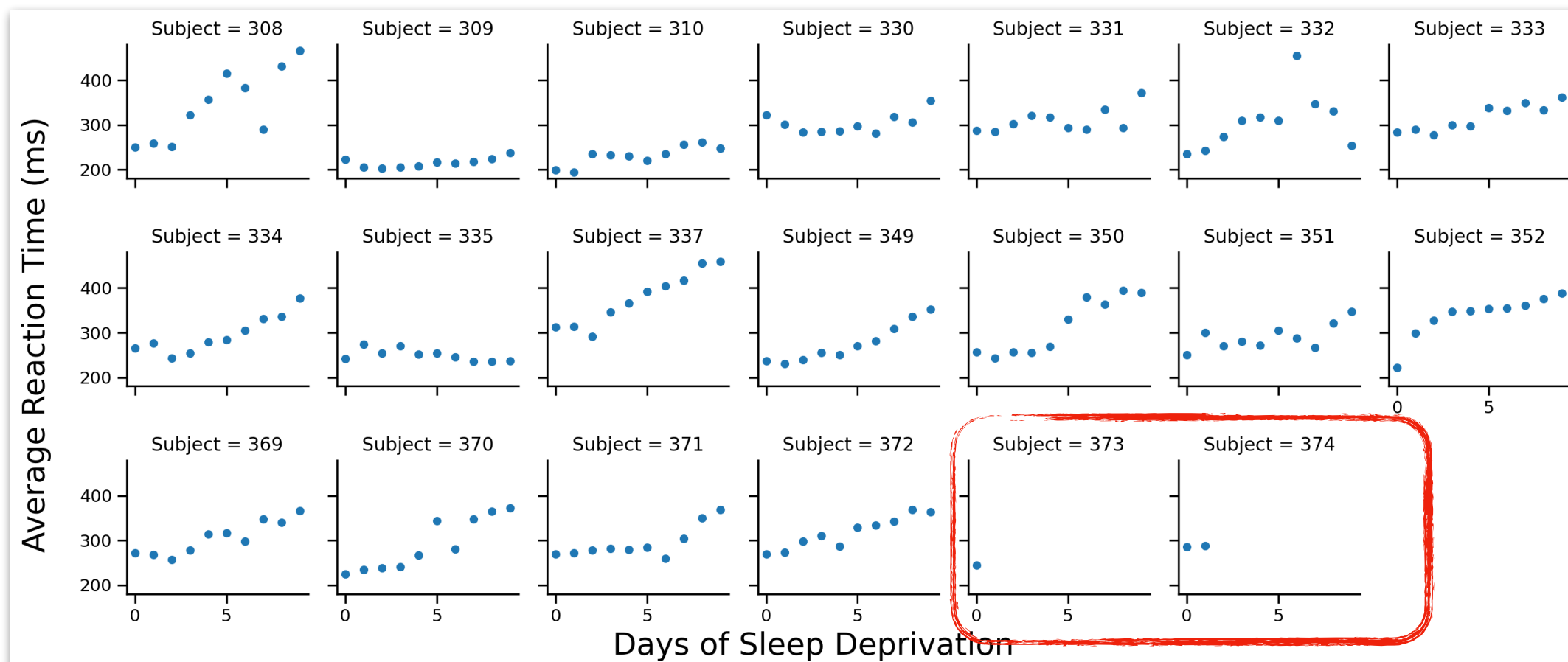


Data set

How does sleep deprivation affect reaction time?

20 participants; 5 measurements per participant

subject	days	reaction
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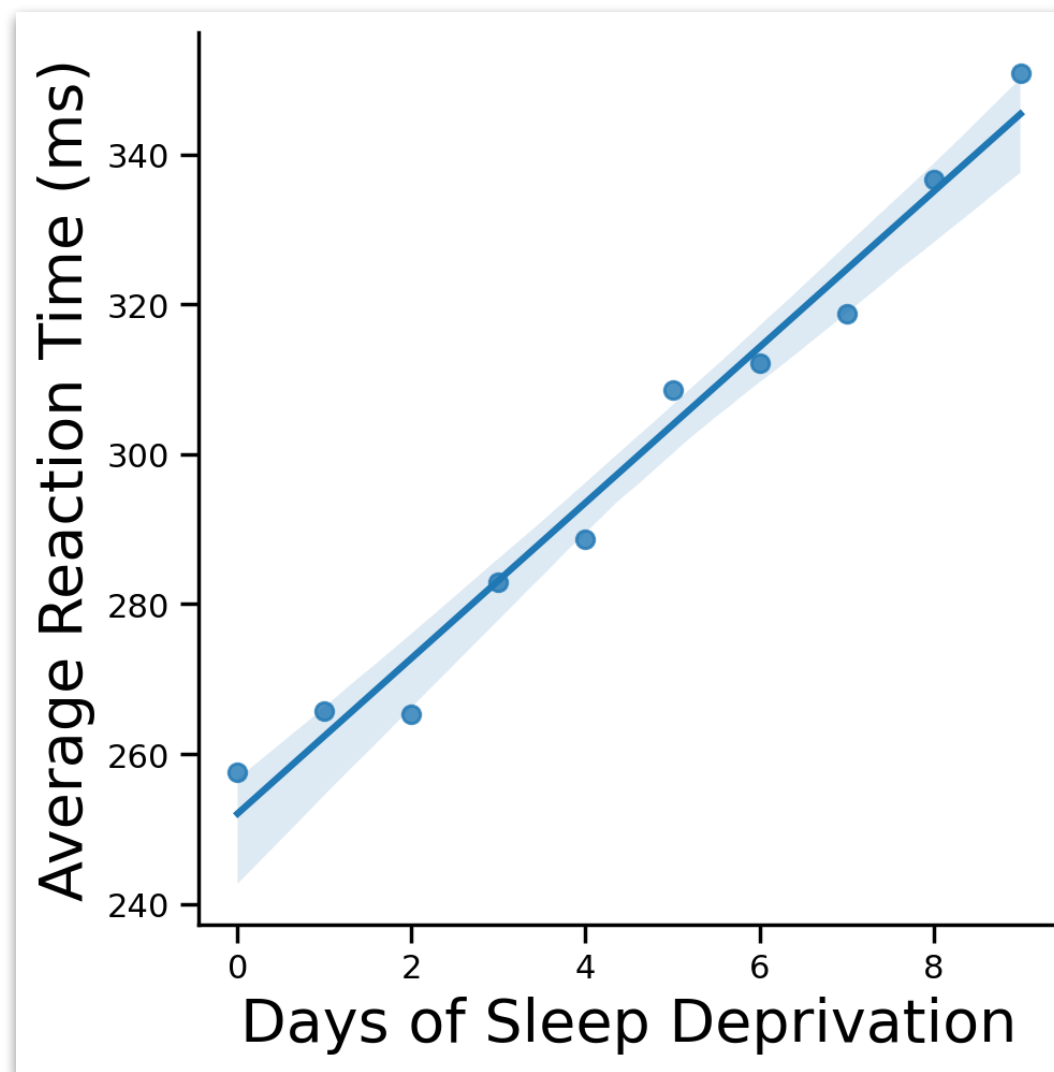
2 with incomplete information

Ways to Aggregate dependent data

Averaging over replicates: average participants within day and run a regression on the averages

Avoid this whenever possible

subject	days	reaction
308	0	249.56
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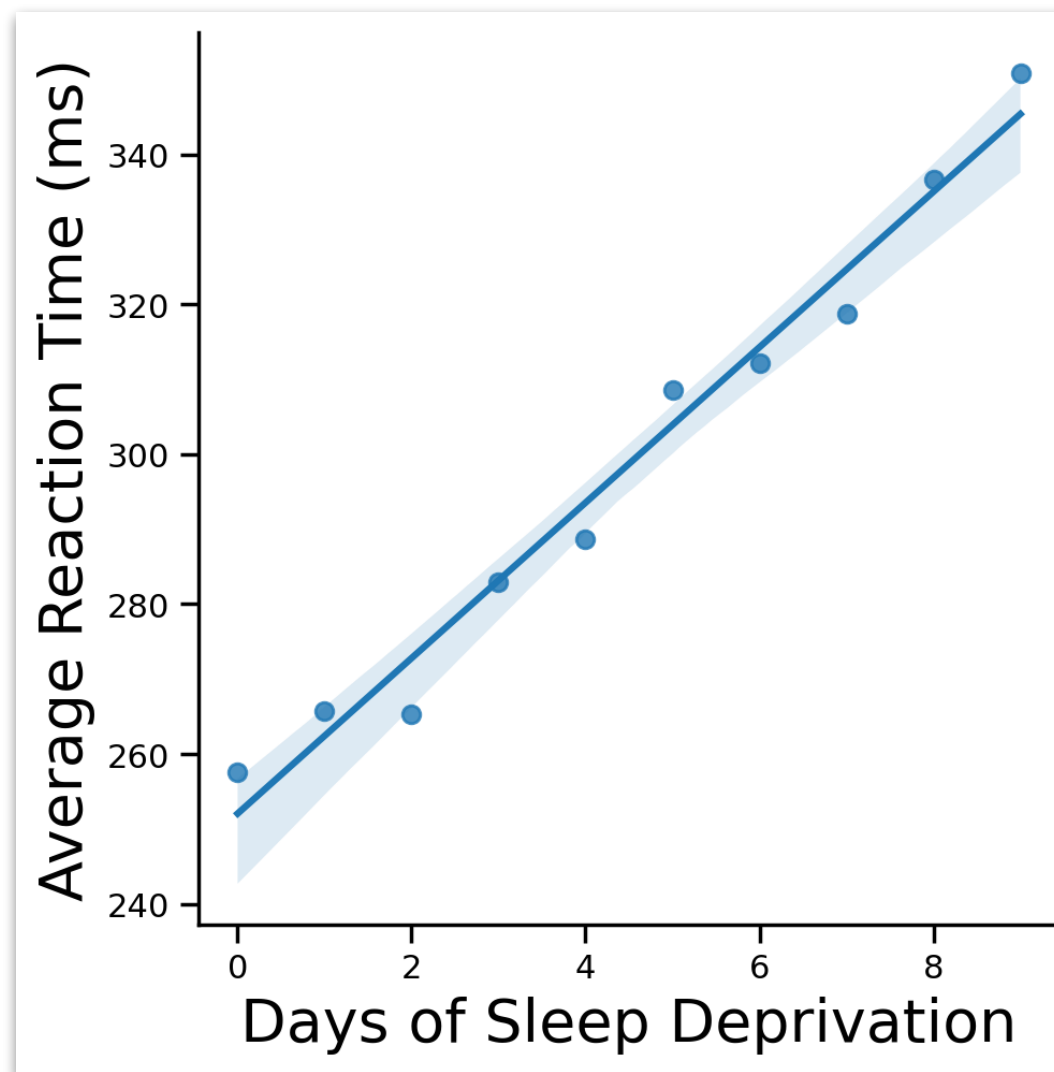
Unnecessarily throws away useful information about variance between people!

Ways to Aggregate dependent data

Averaging over replicates: average participants within day and run a regression on the averages

Avoid this whenever possible

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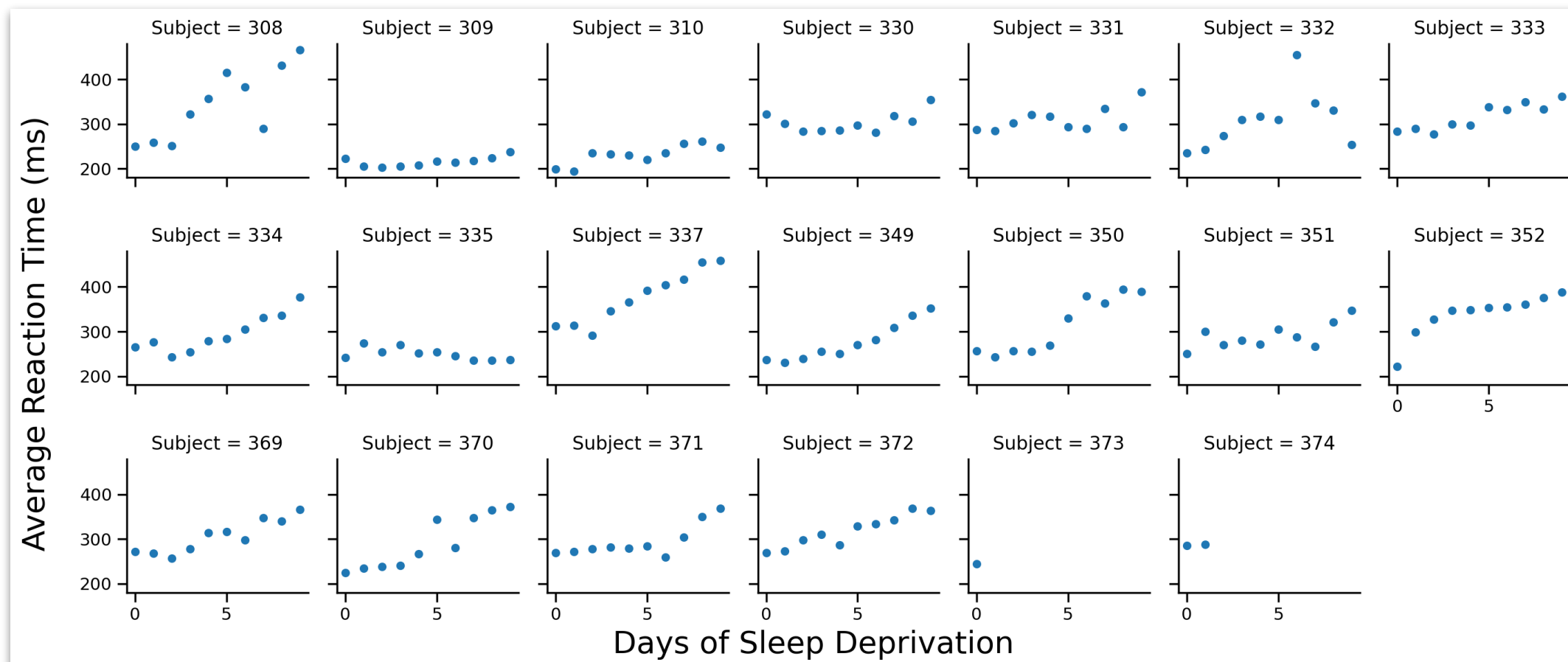
Ok we did do this...for cats HW by averaging over *trials* within each cat...but just to simplify things!

Ways to Aggregate dependent data

1) **Complete pooling**: combine data from all participants (ignore dependence) and fit one **global regression**

20 participants; 5 measurements per participant

subject	days	reaction
308	0	249.56
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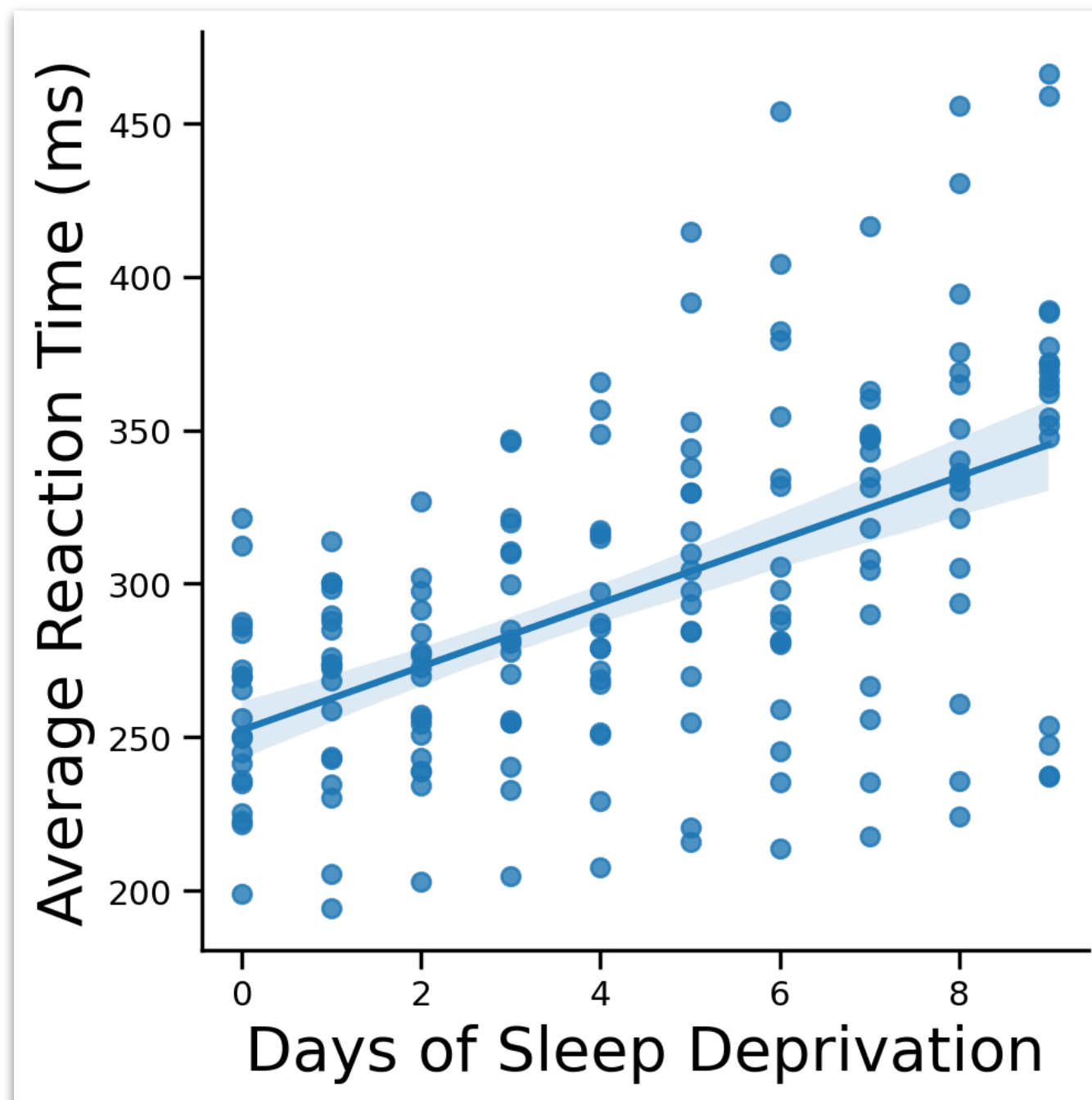


2 with incomplete information

Ways to Aggregate dependent data

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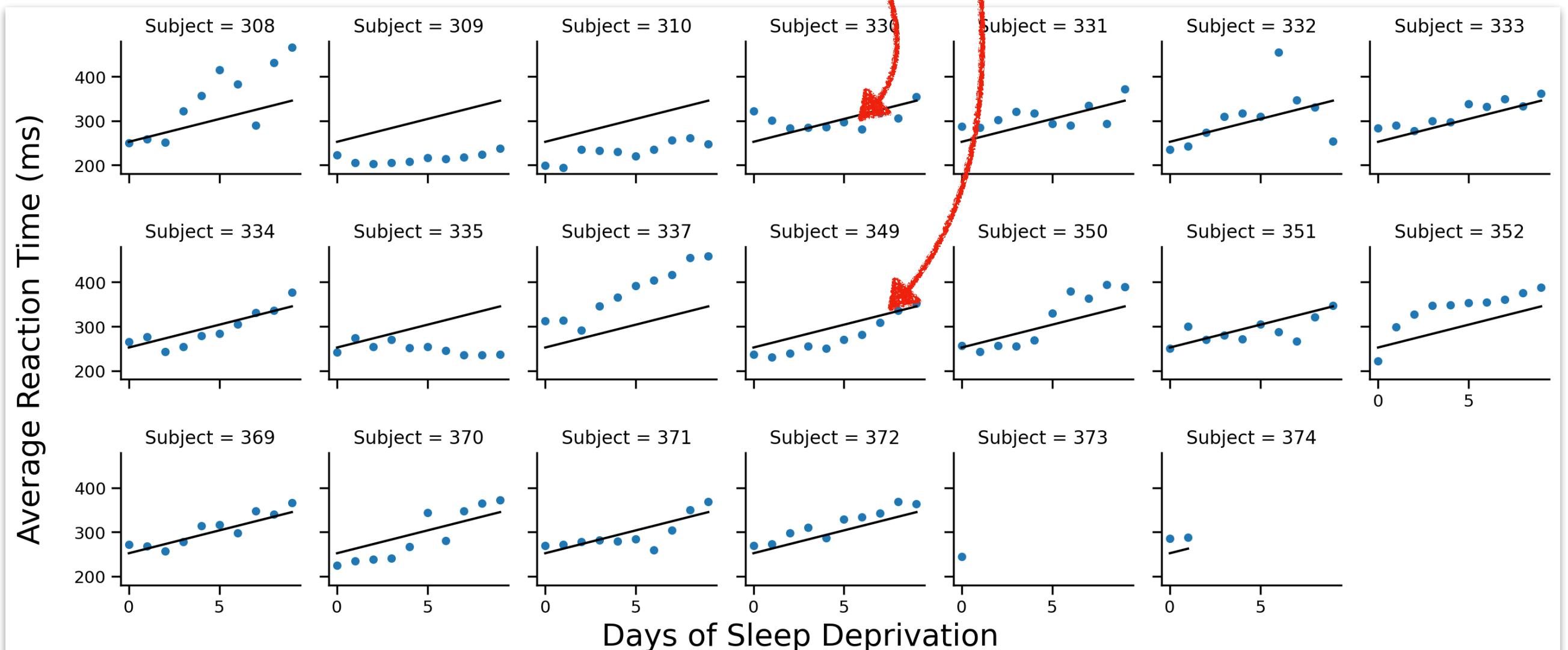
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Complete pooling: Fit one global regression

```
ols("Reaction ~ Days", data = df.to_pandas())
```

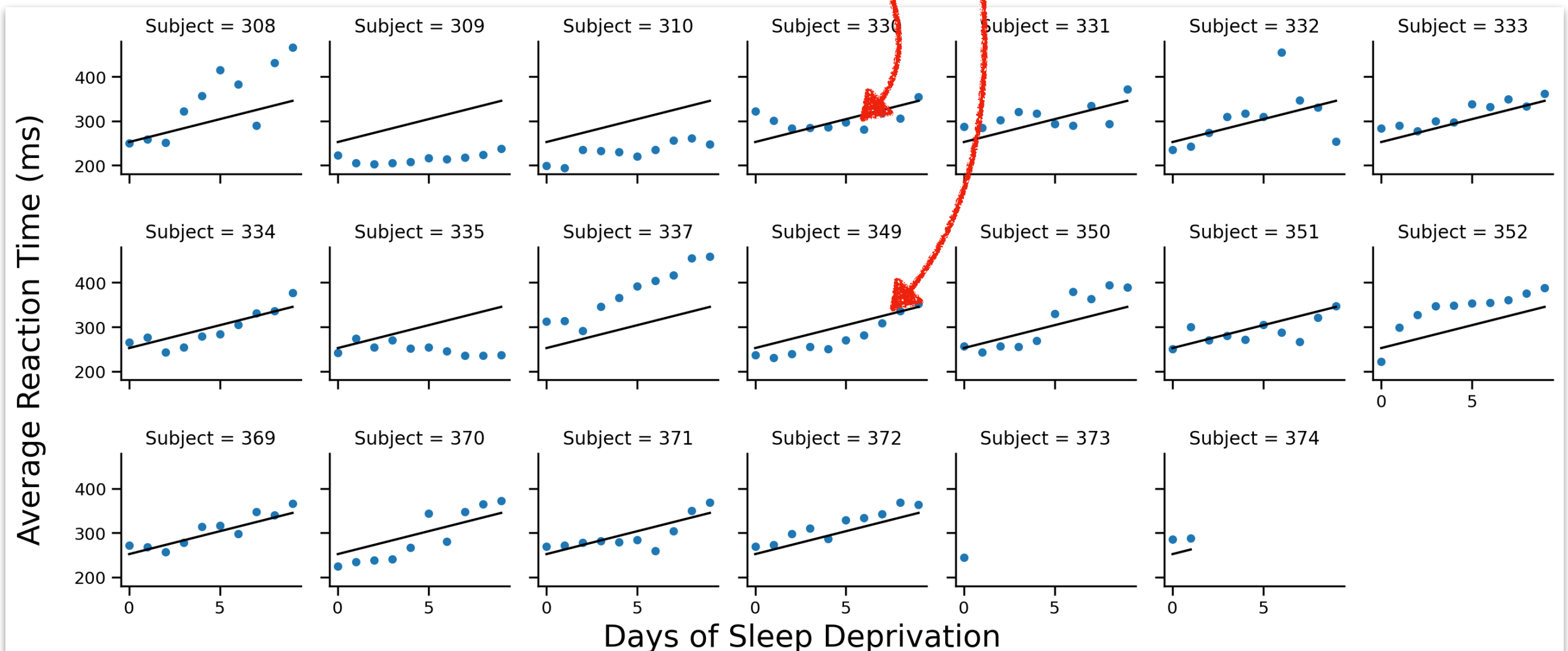
Same regression intercept and slope
for every Subject



Complete pooling: Fit one global regression

Ignores variability across Subjects!

Same regression intercept and slope
for every Subject

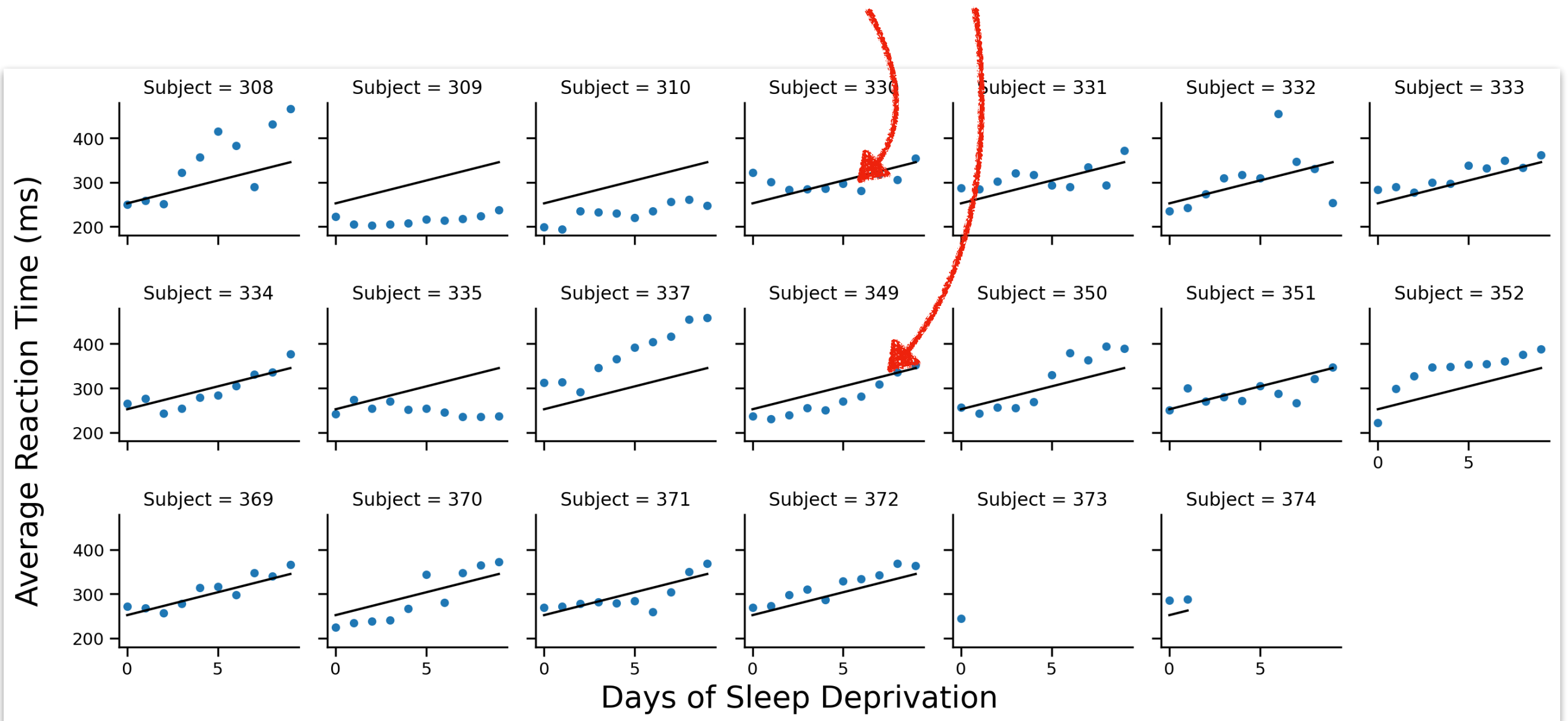


Complete pooling: Fit one global regression

Ignores variability across Subjects!

Risks Simpson's Paradox!

Same regression intercept and slope
for every Subject

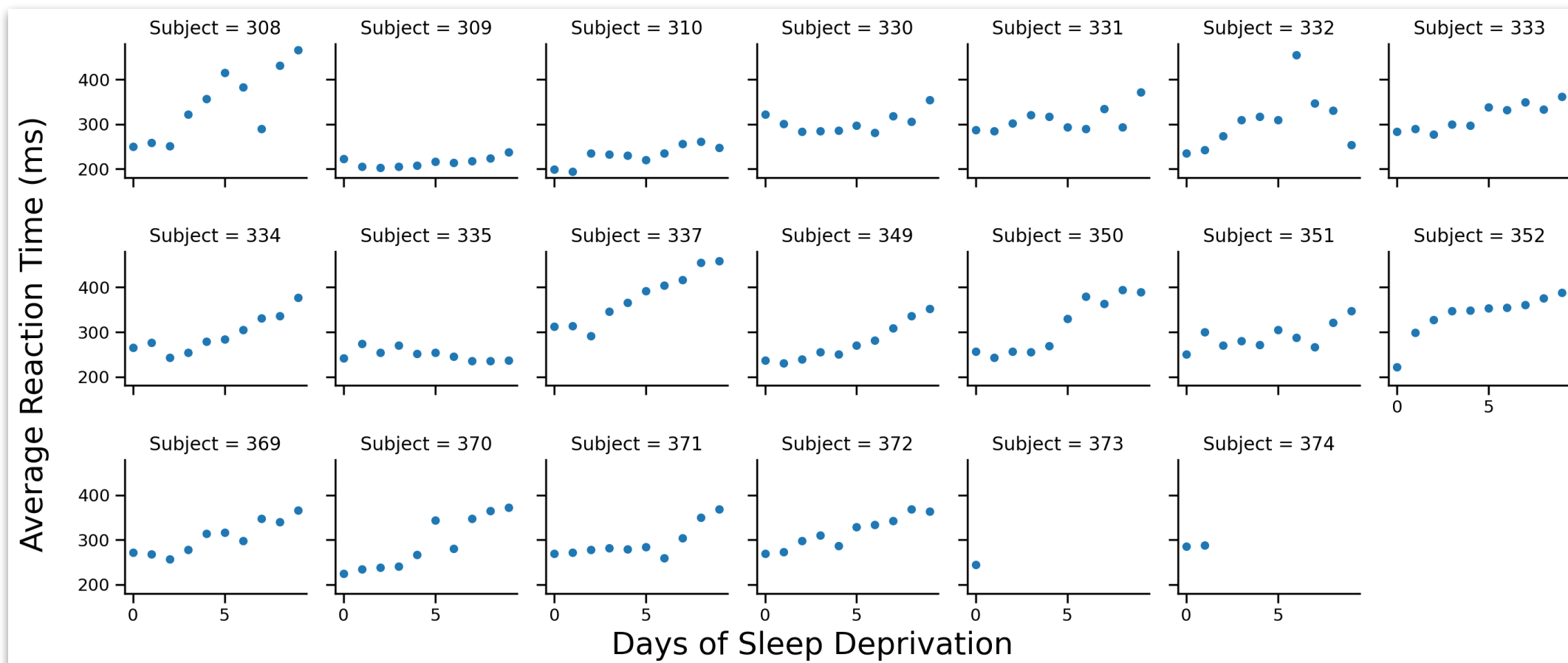


Ways to Aggregate dependent data

2) **No pooling**: don't combine any data and fit 20 separate regressions, one for each Subject

20 participants; 5 measurements per participant

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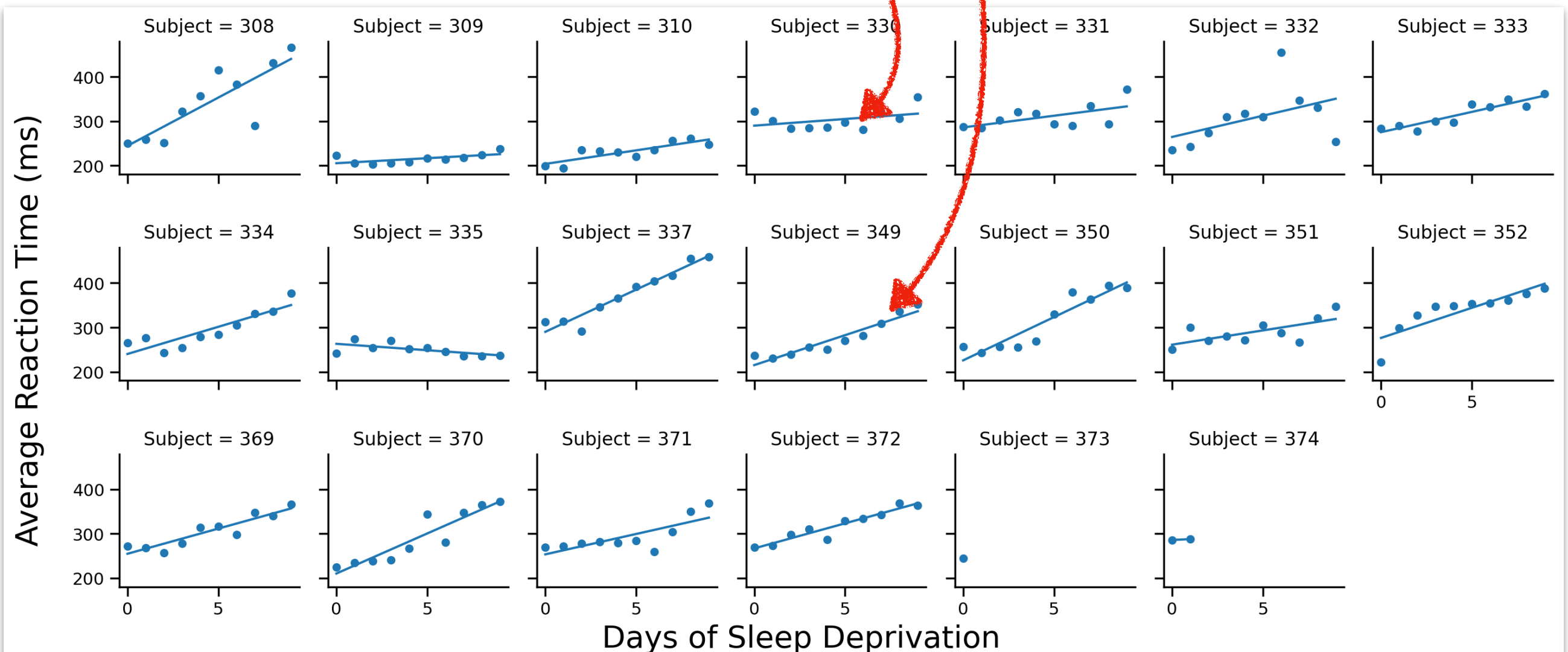


2 with incomplete information

No pooling: Fit separate regressions

```
for subjectID in df["Subject"].unique():  
    sub_data = df.filter(col("Subject") == subjectID)  
    ols("Reaction ~ Days", data = sub_data.to_pandas())
```

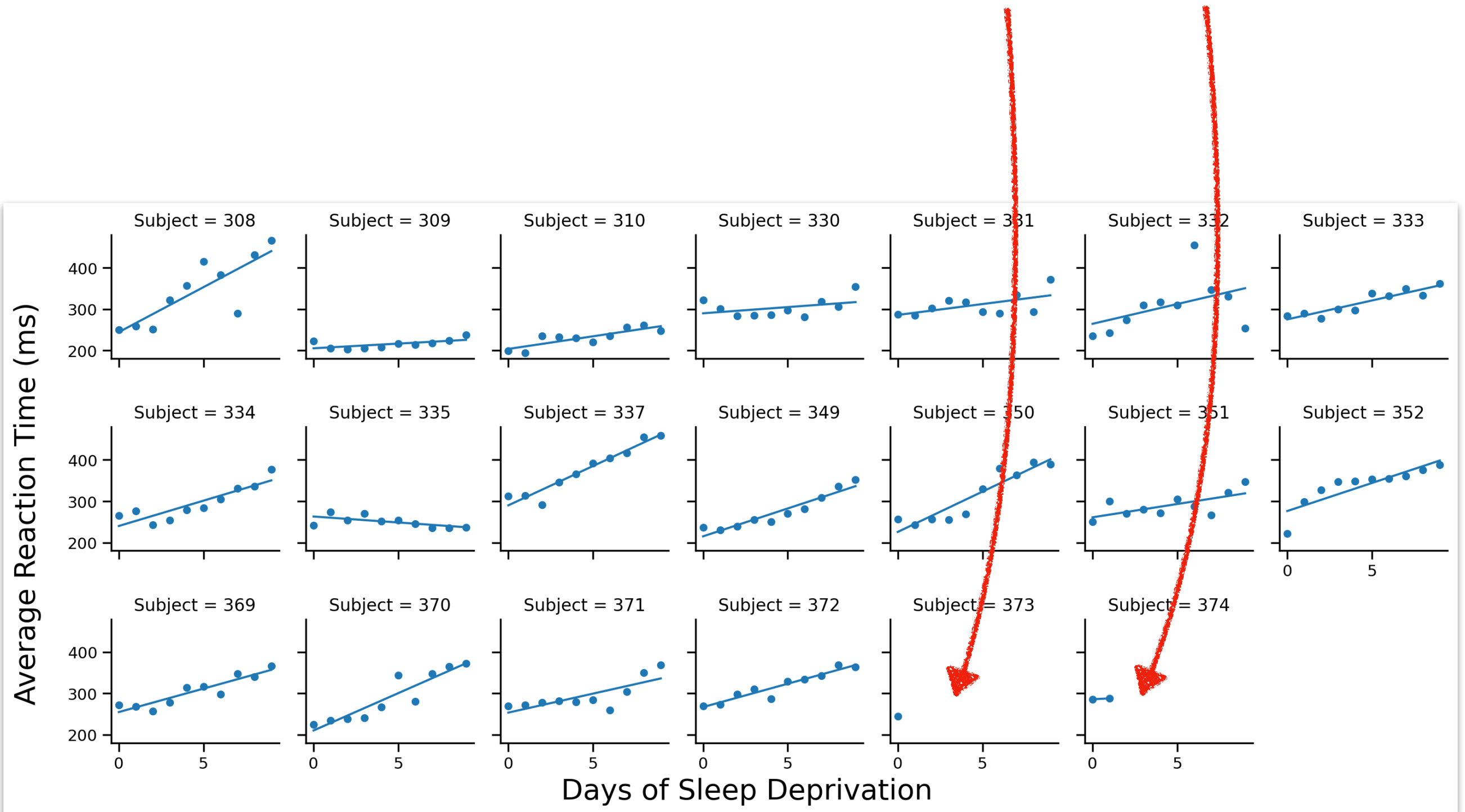
Different regression intercept and slope
for every Subject



No pooling: Fit separate regressions

Treats each subject as a **separate dataset**

Poor/inestimable models when too little/missing data

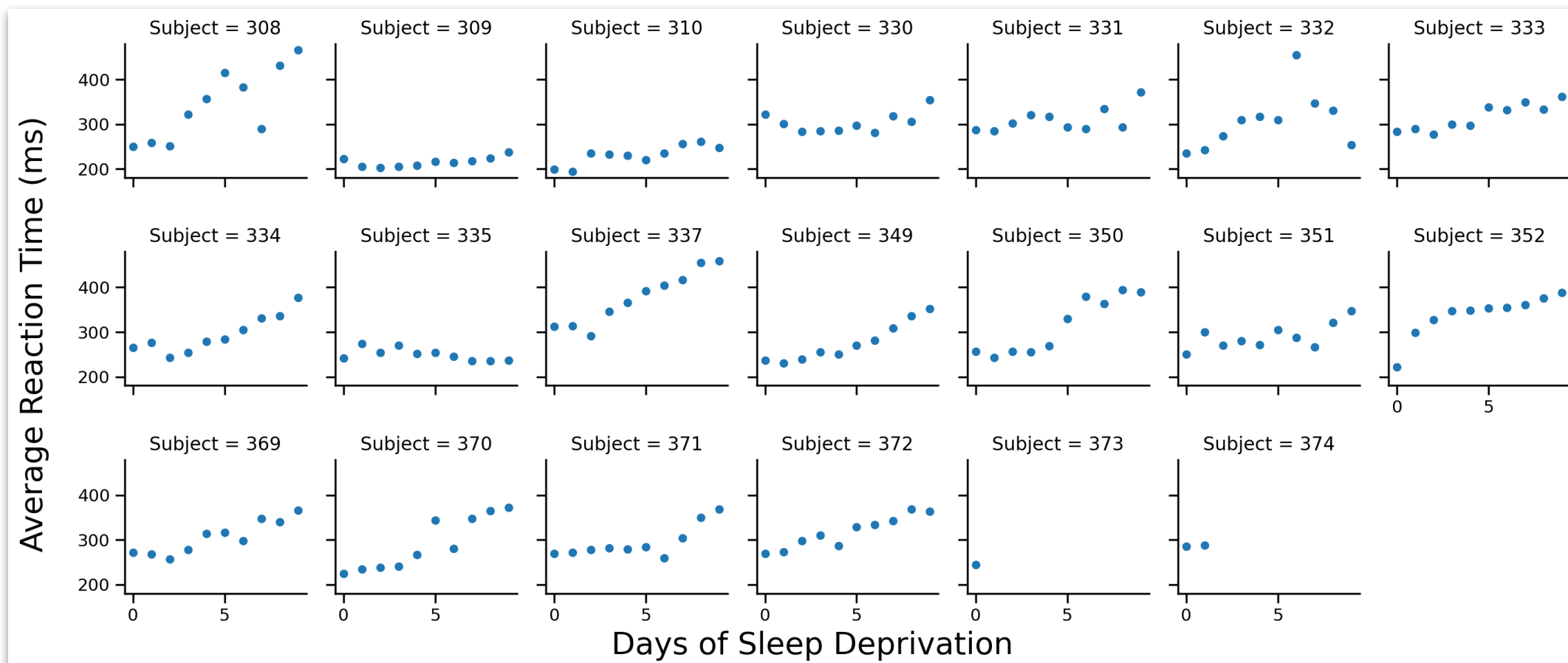


Ways to Aggregate dependent data

3) **Partial pooling**: take into account all data by **explicitly modeling variation** between participants

20 participants; 5 measurements per participant

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2 with incomplete information

Partial pooling: Fit Linear Mixed Model

New Python code alert!

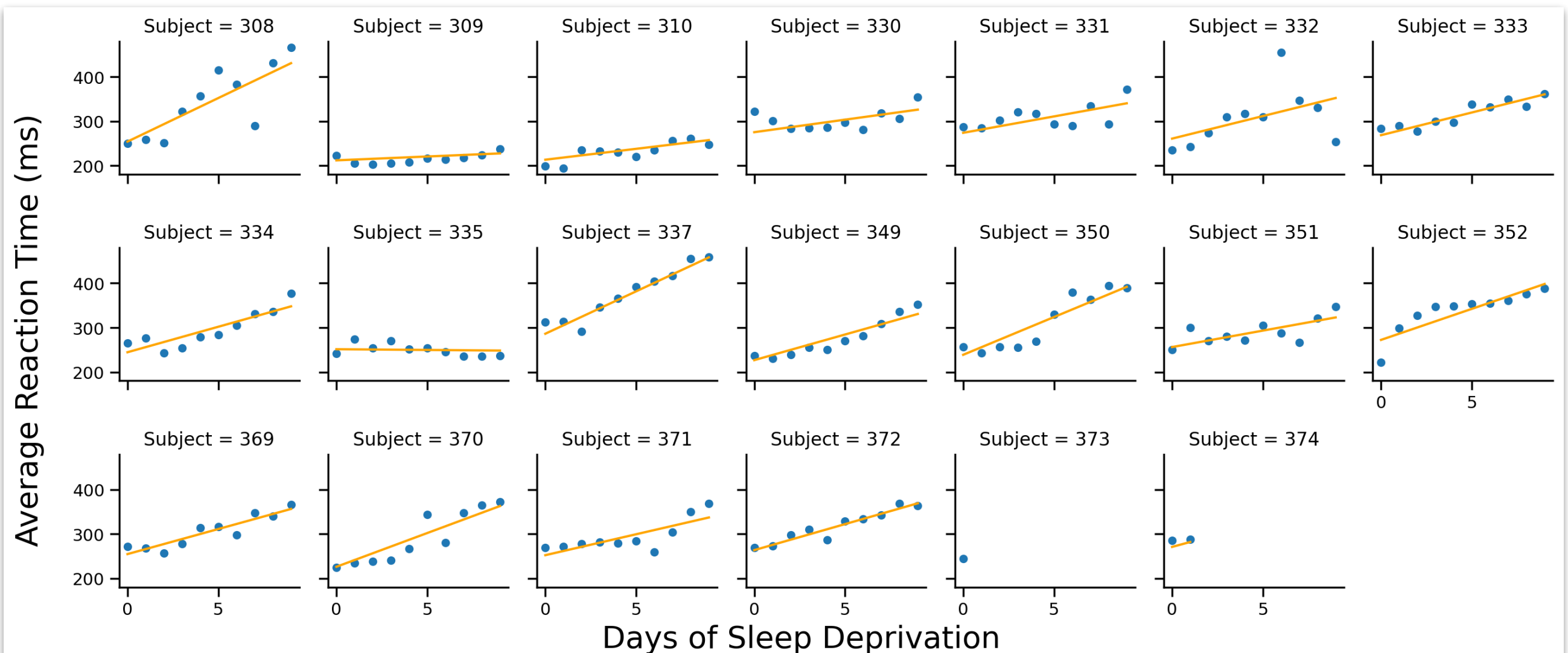
```
Lmer("Reaction ~ Days + (1 + Days | Subject)",  
data = df.to_pandas())
```

intercept

slope

how to model dependence

Each Subject is assumed to have unique **Intercept & Slope**



Ways to Aggregate dependent data

complete pooling

Doesn't account for individual variation

no pooling

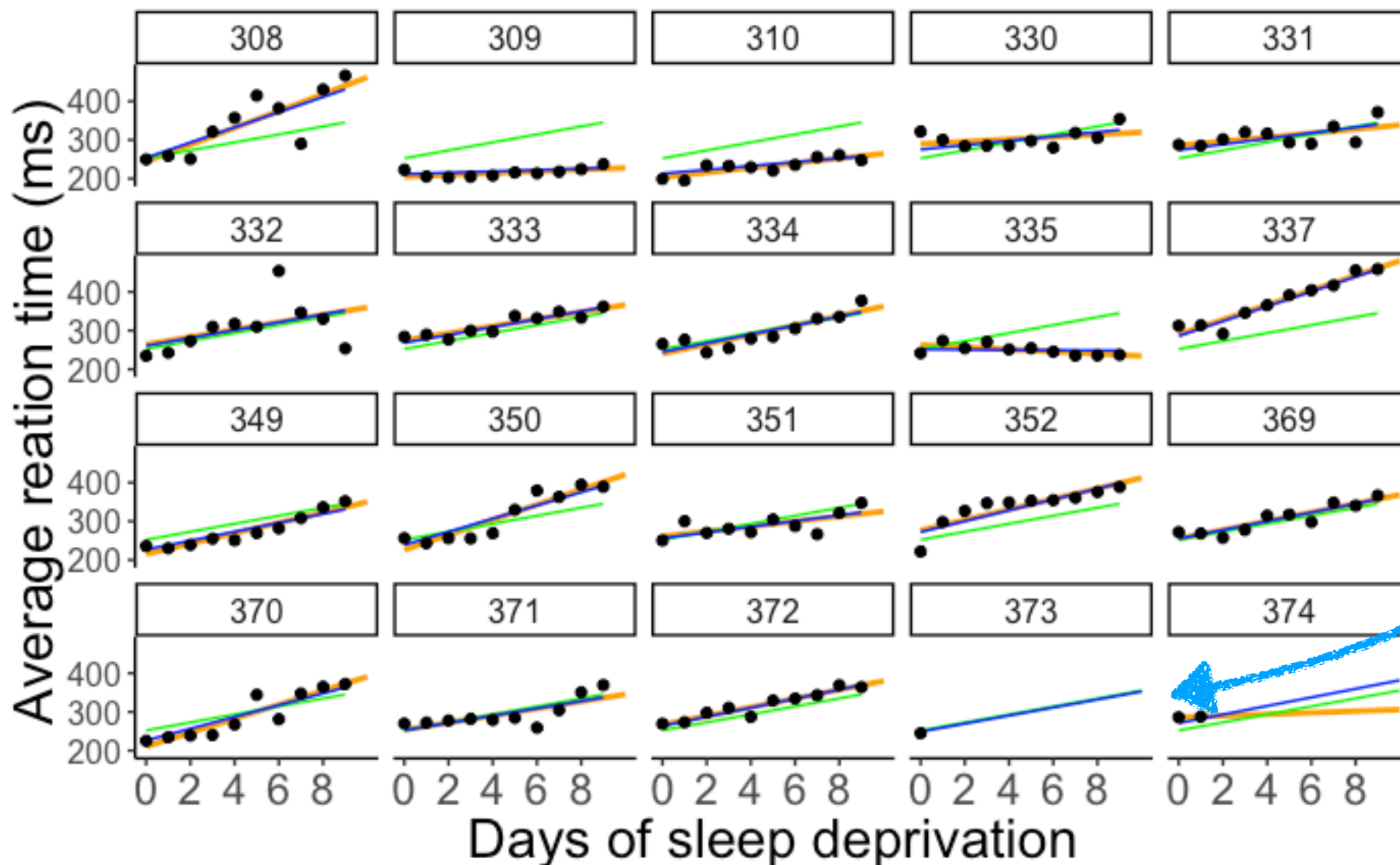
No/noisy predictions with one/few observations

Doesn't account for shared/general effect across Ss

partial pooling
(LMMs)

Explicit model of multiple sources of variation

Extrapolation possible when data is missing



LMMs: The Essentials

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Linear *Mixed* Model: What's getting Mixed?

**Every field agrees these are terrible names*

Fixed Effects/Factors

Random Effects/Factors

- **Why I don't use the term "fixed and random effects"**

Posted on [January 25, 2005 6:41 PM](#) by [Andrew](#)

People are always asking me if I want to use a fixed or random effects model for this or that. I always reply that these terms have no agreed-upon definition. People with their own favorite definition of "fixed and random effects" don't always realize that other definitions are out there. Worse, people conflate different definitions.

our experiment)

participants, stimuli, etc)

- parameters are the *same* across repeated measures
- sometimes call *population average effect*

- parameters *vary* for each "cluster" of repeated measures
- provides explicit model of dependence in the data

Linear *Mixed* Model: What's getting Mixed?

The inference we want to make about the population from our sample

Fixed Effects/Factors

- parameter estimates we typically care about just like normal regression (e.g. IVs in our experiment)
- parameters are the *same* across repeated measures
- sometimes call *population average effect*

The variation/uncertainty we want to generalize over when making that inference

Random Effects/Factors

- the *variation* we want to take into account (e.g. differences between departments, participants, stimuli, etc)
- parameters *vary* for each "cluster" of repeated measures
- provides explicit model of dependence in the data

OLS Regression only estimates these!

The inference we want to make about the population from our sample

Fixed Effects/Factors

- parameter estimates we typically care about just like normal regression (e.g. IVs in our experiment)
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Random Effects/Factors

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Linear *Mixed* Model: What's getting Mixed?

The inference we want to make about the population from our sample

The variation/uncertainty we want to generalize over when making that inference

Fixed Effects/Factors

Random Effects/Factors

"Reaction ~ Days + (1 + Days | Subject)"

What is the relationship between **Days** and **Reaction** in population?

subject	days	reaction
308	0	249.56
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Accounting for the fact that each **Subject** can have a different baseline **Reaction** and different effect of **Days**...

Linear *Mixed* Model: What's getting Mixed?

The inference we want to make about the population from our sample

The variation/uncertainty we want to generalize over when making that inference

Fixed Effects/Factors

Random Effects/Factors

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What is the relationship between Days and Reaction in population?

Accounting for the fact that each **Subject** can have a different baseline **Reaction** and different effect of **Days**...

Estimates inform each other and result in **shrinkage**

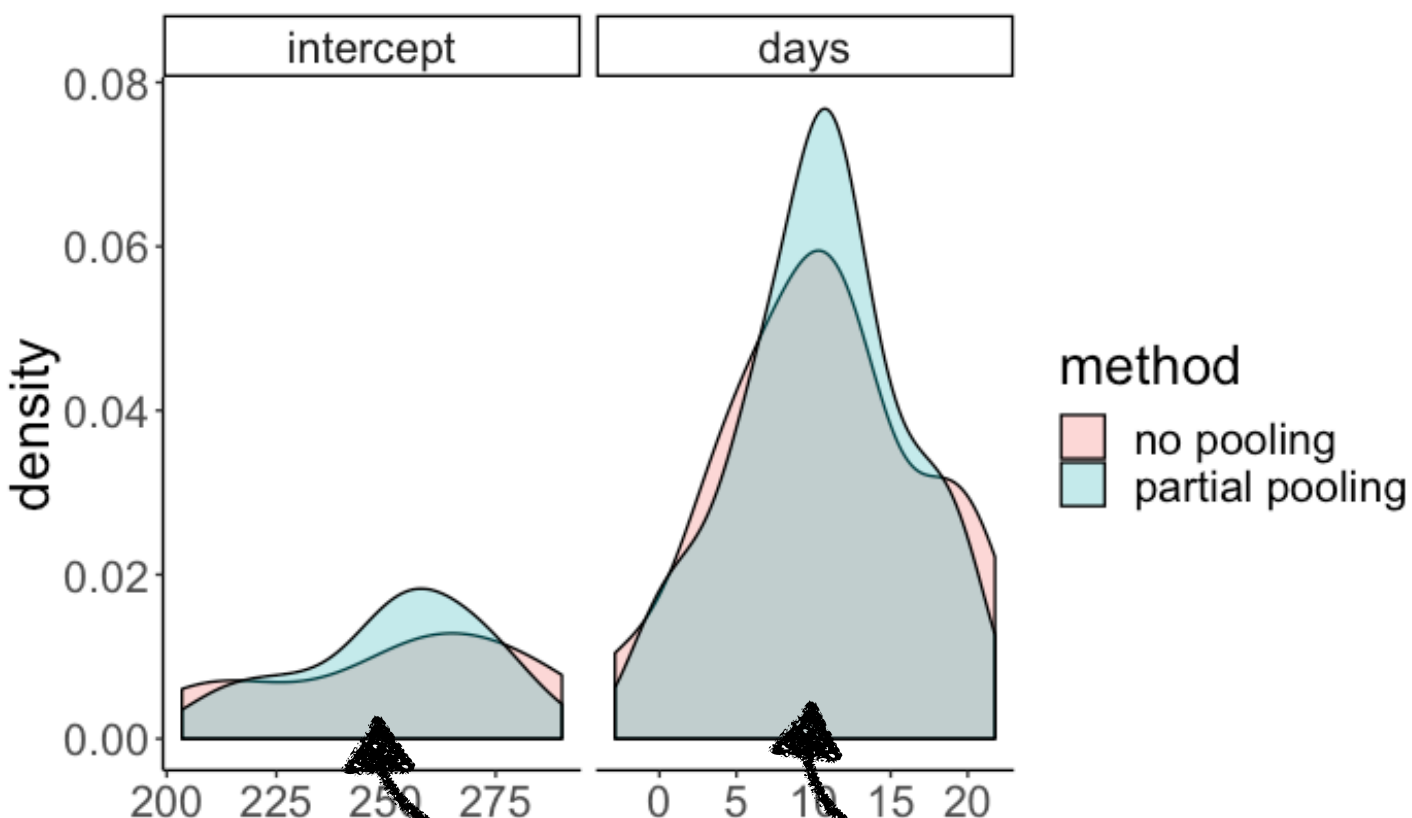
Linear *Mixed* Model: What's getting Mixed?

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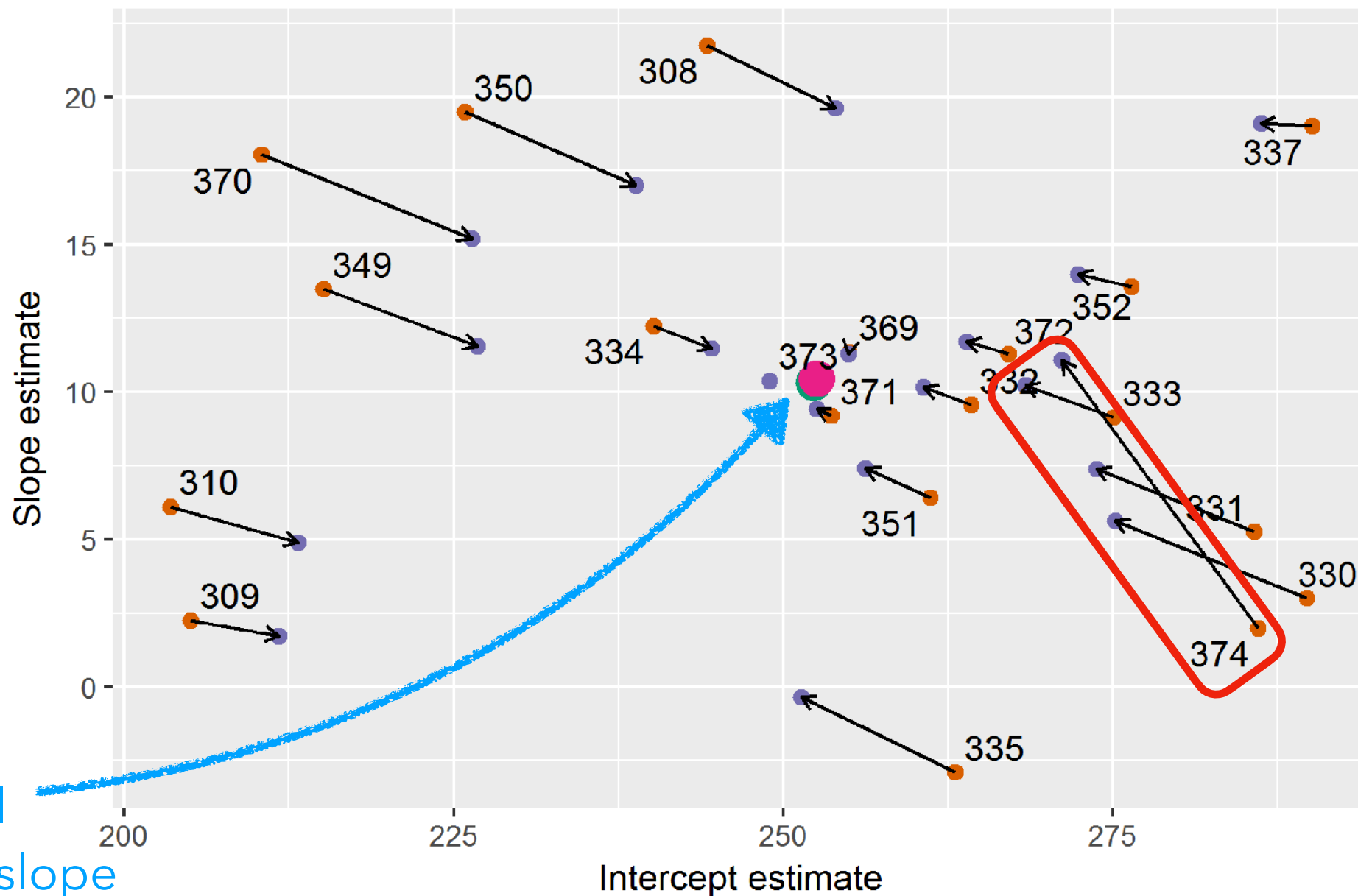


Variance "shrinks"

method	intercept	days
no pooling	28.95	6.56
partial pooling	21.59	5.46

Distributions over Subject

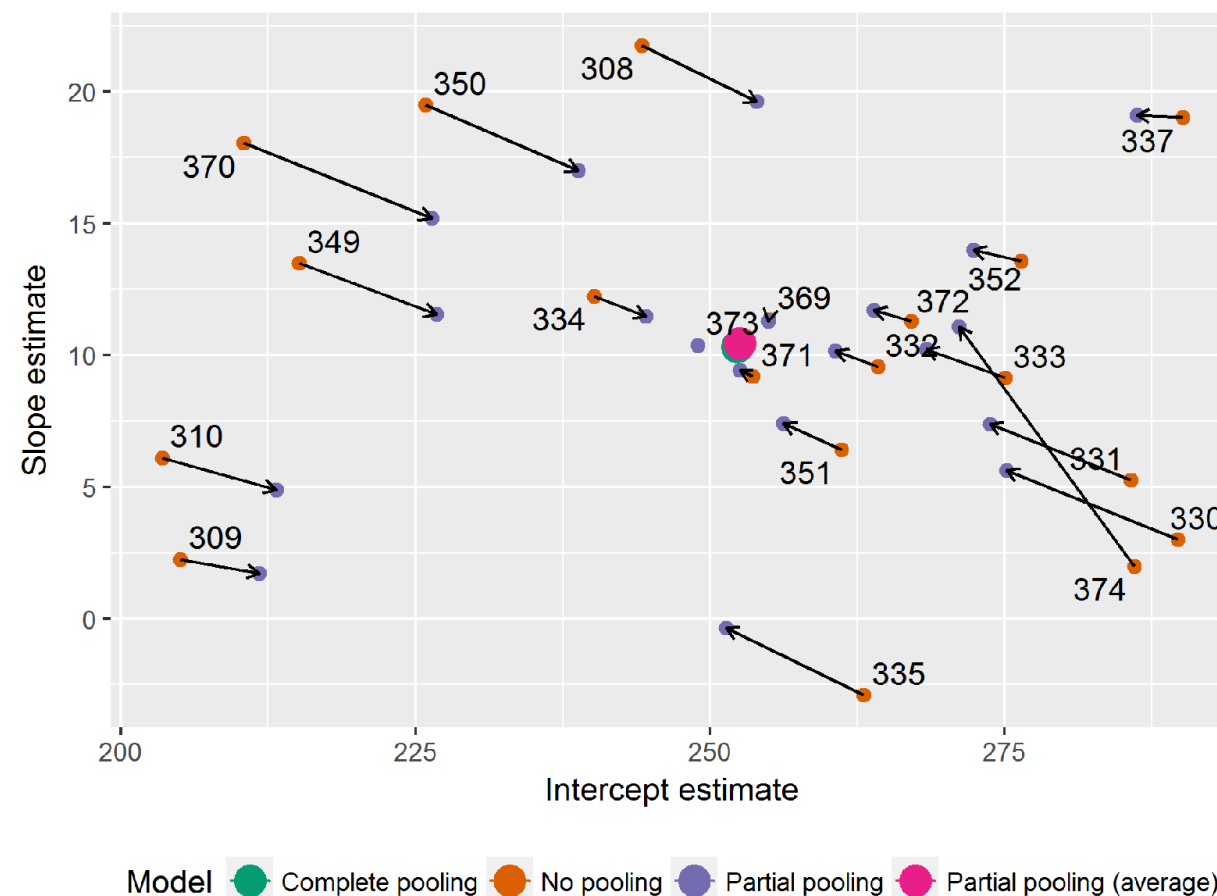
Estimates inform each other and result in **shrinkage**



we only had
two
observations

- More shrinkage when estimate is further from the average
- More shrinkage when estimate is more uncertain (based on fewer observations); more information "borrowed" from other clusters

Estimates inform each other and result in **shrinkage**



The term “shrinkage” may have negative connotations. John Tukey preferred to refer to the process as the estimates for individual subjects “**borrowing strength**” from each other.

This is a **fundamental difference between mixed-effects models versus strictly fixed effects models**. In a mixed-effects model we assume that the levels of a grouping factor are a selection from a population and, as a result, can be expected to share characteristics to some degree. Consequently, the predictions from a mixed-effects model are attenuated relative to those from strictly fixed-effects models.

Douglas Bates (author of LMM library in R lme4)

Linear Mixed Model: What's happening under-the-hood?

Regular Regression (OLS) `ols()`

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$\text{Reaction}_i = \beta_0 + \beta_1 \text{Days}_i + \epsilon_i$$

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
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309	0	222.73
309	1	205.27
309	2	202.98

i = single observation/row

Linear Mixed Model: What's happening under-the-hood?

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$$Reaction_i = \beta_0 + \beta_1 Days_i + \epsilon_i$$

$i =$ single observation/row

Fixed effects

Linear Mixed Model (LMM) `Lmer()`

$$y_{ij} = \underline{(\beta_0 + b_{0j})} + \underline{(\beta_1 + b_{1j})}X_{ij} + \epsilon_{ij}$$

Linear Mixed Model: What's happening under-the-hood?

Regular Regression (OLS) `ols()`

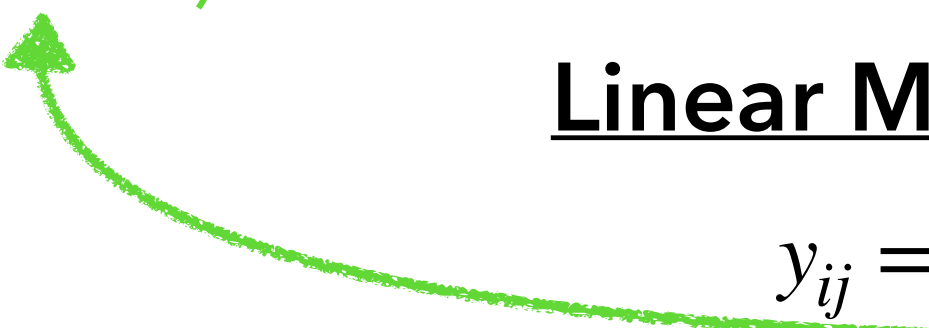
$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$Reaction_i = \beta_0 + \beta_1 Days_i + \epsilon_i$$

$i =$ single observation/row

Random effects
(offsets)

Linear Mixed Model (LMM) `Lmer()`

$$y_{ij} = (\beta_0 + \underline{b_{0j}}) + (\beta_1 + \underline{b_{1j}})X_{ij} + \epsilon_{ij}$$


Linear Mixed Model: What's happening under-the-hood?

Regular Regression (OLS) `ols()`

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ij = i observation for subject j

Linear Mixed Model: What's happening under-the-hood?

Regular Regression (OLS) `ols()`

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$$\text{Reaction}_i = \beta_0 + \beta_1 \text{Days}_i + \epsilon_i$$

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Linear Mixed Model (LMM) `Lmer()`

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$$\text{Reaction}_{ij} = (\beta_0 + \underline{b_{0j}}) + (\beta_1 + \underline{b_{1j}})\text{Days}_{ij} + \epsilon_{ij}$$

Subject j 's
intercept & slope

$ij =$ i observation for subject j

Linear Mixed Model: What's happening under-the-hood?

Linear Mixed Model (LMM) `lmer()`

$$y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{ij} + \epsilon_{ij}$$

$$Reaction_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})Days_{ij} + \epsilon_{ij}$$

Really 2 equations (levels)

General relationship (fixed) $Y_{sd} = \beta_{0s} + \beta_{1s}X_{sd} + e_{sd}$ Level 1

Participant level (random) $\beta_{0s} = \gamma_0 + S_{0s}$
 $\beta_{1s} = \gamma_1 + S_{1s}$ Level 2

Variance across Intercepts
Variance across Slopes

$$\langle S_{0s}, S_{1s} \rangle \sim N(\langle 0, 0 \rangle, \Sigma)$$

$$\Sigma = \begin{pmatrix} \tau_{00}^2 & \rho\tau_{00}\tau_{11} \\ \rho\tau_{00}\tau_{11} & \tau_{11}^2 \end{pmatrix}$$

Error for each obs for each S

$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

Linear Mixed Model (LMM) `lmer()`

$$Reaction_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})Days_{ij} + \epsilon_{ij}$$

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
X_{sd}	observed	Value of days_deprived (0-7) for subject s on day d
β_{0s}	derived	level 1 intercept parameter
β_{1s}	derived	level 1 slope parameter
e_{sd}	derived	Error for subject s , day d
γ_0	fixed	Grand intercept ("gamma")
γ_1	fixed	Grand slope ("gamma")
S_{0s}	derived	Random intercept (offset) for subject s
S_{1s}	derived	Random slope (offset) for subject s
Σ	random	Variance-covariance matrix
τ_{00}^2	random	Variance of random intercepts
ρ	random	Random correlation between intercepts and slopes
τ_{11}^2	random	Variance of random slopes
σ^2	random	Error variance

Level 1

$$Y_{sd} = \beta_{0s} + \beta_{1s}X_{sd} + e_{sd}$$

Level 2

$$\beta_{0s} = \gamma_0 + S_{0s}$$

$$\beta_{1s} = \gamma_1 + S_{1s}$$

Variance Components

$$\langle S_{0s}, S_{1s} \rangle \sim N(\langle 0,0 \rangle, \Sigma)$$

$$\Sigma = \begin{pmatrix} \tau_{00}^2 & \rho\tau_{00}\tau_{11} \\ \rho\tau_{00}\tau_{11} & \tau_{11}^2 \end{pmatrix}$$

$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate **fixed effects** from **data**

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
X_{sd}	observed	Value of days_deprived (0-7) for subject s on day d
β_{0s}	derived	level 1 intercept parameter
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Level 2

$$\beta_{0s} = \gamma_0 + S_{0s}$$

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Variance Components

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$$\Sigma = \begin{pmatrix} \tau_{00}^2 & \rho\tau_{00}\tau_{11} \\ \rho\tau_{00}\tau_{11} & \tau_{11}^2 \end{pmatrix}$$

$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate fixed effects from data
- We estimate **variance & covariance** of **random-effects**

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
X_{sd}	observed	Value of days_deprived (0-7) for subject s on day d
β_{0s}	derived	level 1 intercept parameter
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$$Y_{sd} = \beta_{0s} + \beta_{1s}X_{sd} + e_{sd}$$

Level 2

$$\beta_{0s} = \gamma_0 + S_{0s}$$

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Variance Components

$$\langle S_{0s}, S_{1s} \rangle \sim N(\langle 0, 0 \rangle, \Sigma)$$

$$\Sigma = \begin{pmatrix} \tau_{00}^2 & \rho\tau_{00}\tau_{11} \\ \rho\tau_{00}\tau_{11} & \tau_{11}^2 \end{pmatrix}$$

$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate fixed effects from data
- We estimate variance & covariance of random-effects
- Then we can derive (not estimate) Subject offsets from fixed effects

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
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$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate fixed effects from data
- We estimate variance & covariance of random-effects
- Then we can derive (not estimate) Subject offsets from fixed effects

Random effects are the variability around **Fixed effects**

This is *how* LMMs **explicitly model dependence** in data

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
X_{sd}	observed	Value of days_deprived (0-7) for subject s on day d
β_{0s}	derived	level 1 intercept parameter
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LMMs: The Essentials

1. Ways to aggregate dependent data
 - Complete, No-pooling, **Partial-pooling**
2. Parts of the mixed model
 - Fixed vs Random Effects
 - What's happening under-the-hood
3. Specifying Random Effects
 - Random Intercepts & Slopes
 - How to choose your random-effects
4. How to estimate LMMs
 - Meet Pymer4

Specifying Random Effects

New Python code alert!

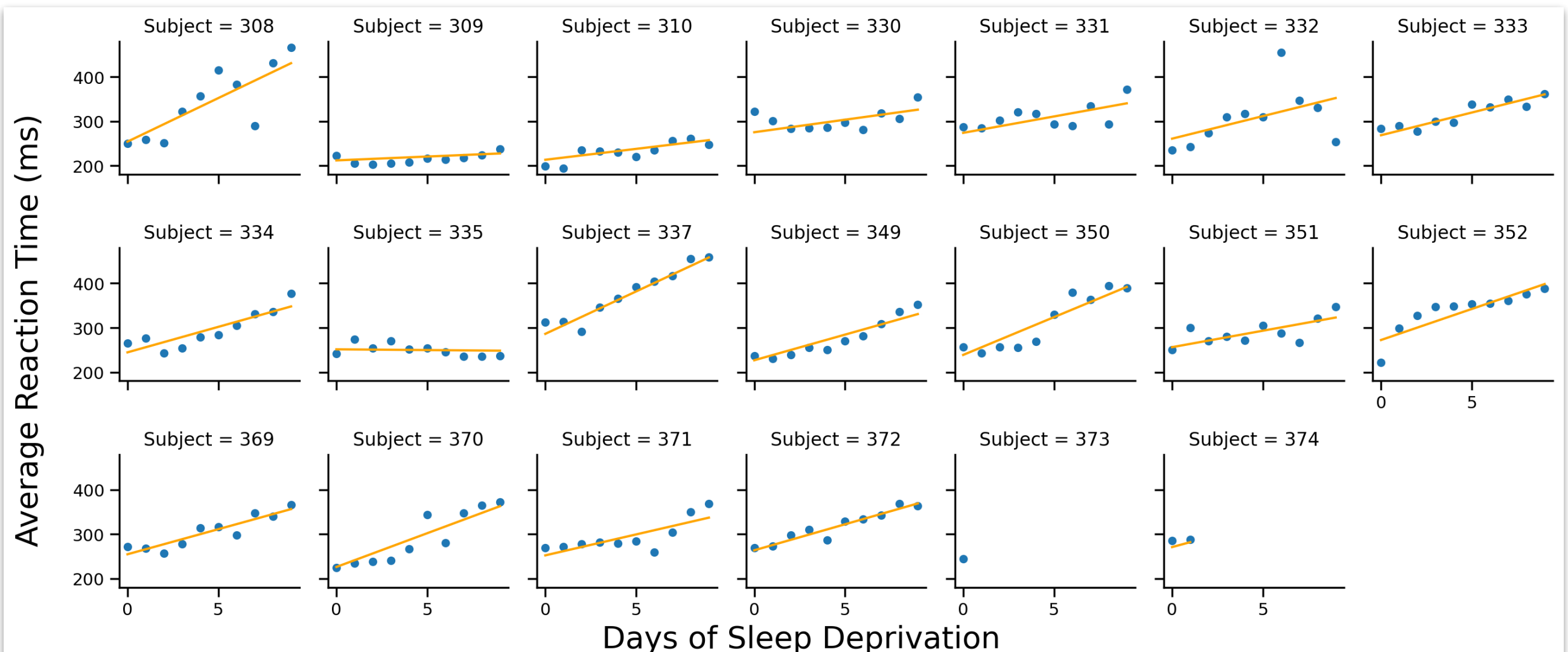
intercept

slope

```
Lmer ("Reaction ~ Days + (1 + Days | Subject)",  
data = df.to_pandas())
```

how to model dependence

Each Subject is assumed to have unique **Intercept & Slope**



Specifying Random Effects

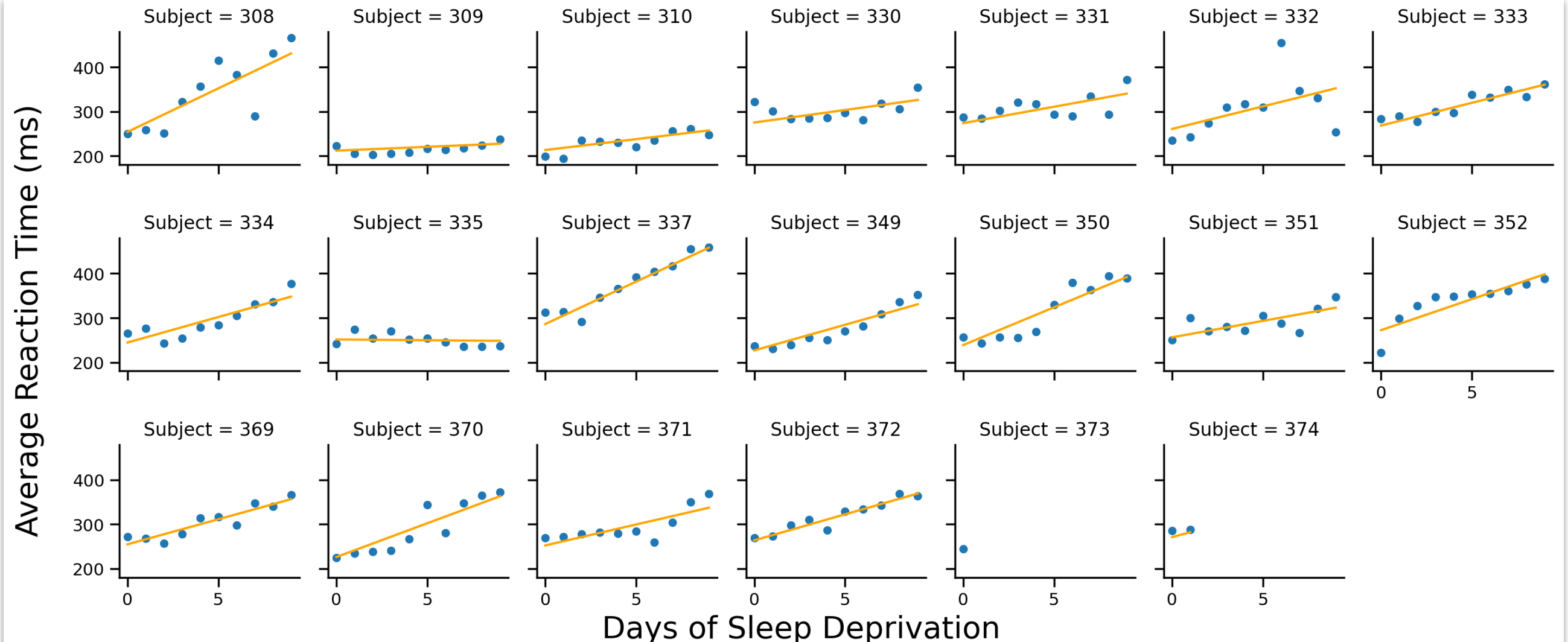
But we can model dependence in **multiple ways**

Each Subject is assumed to have unique **Intercept & Slope**

Varving intercepts

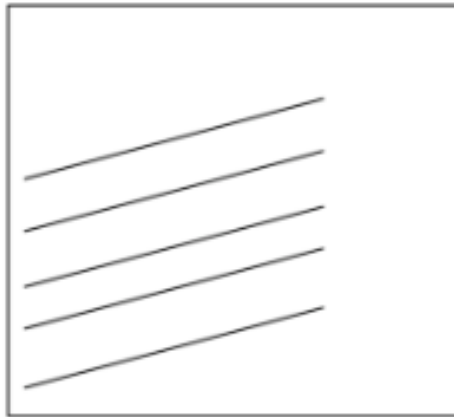
Varving slopes

Varving intercepts and slopes

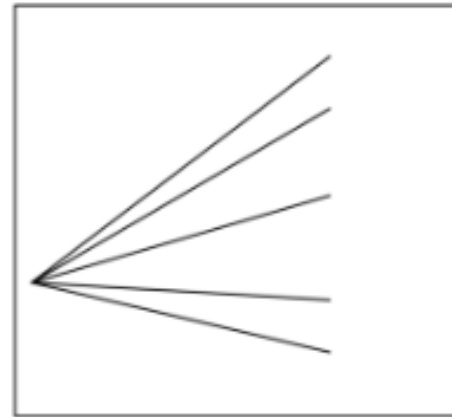


Specifying Random Effects

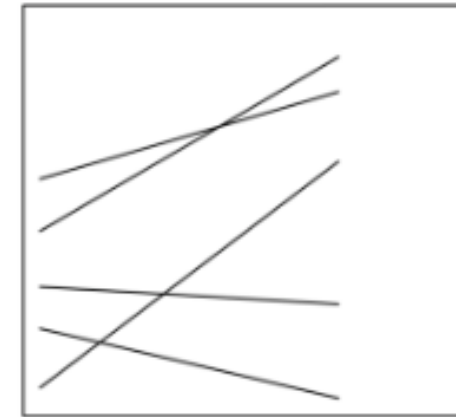
Varying intercepts



Varying slopes



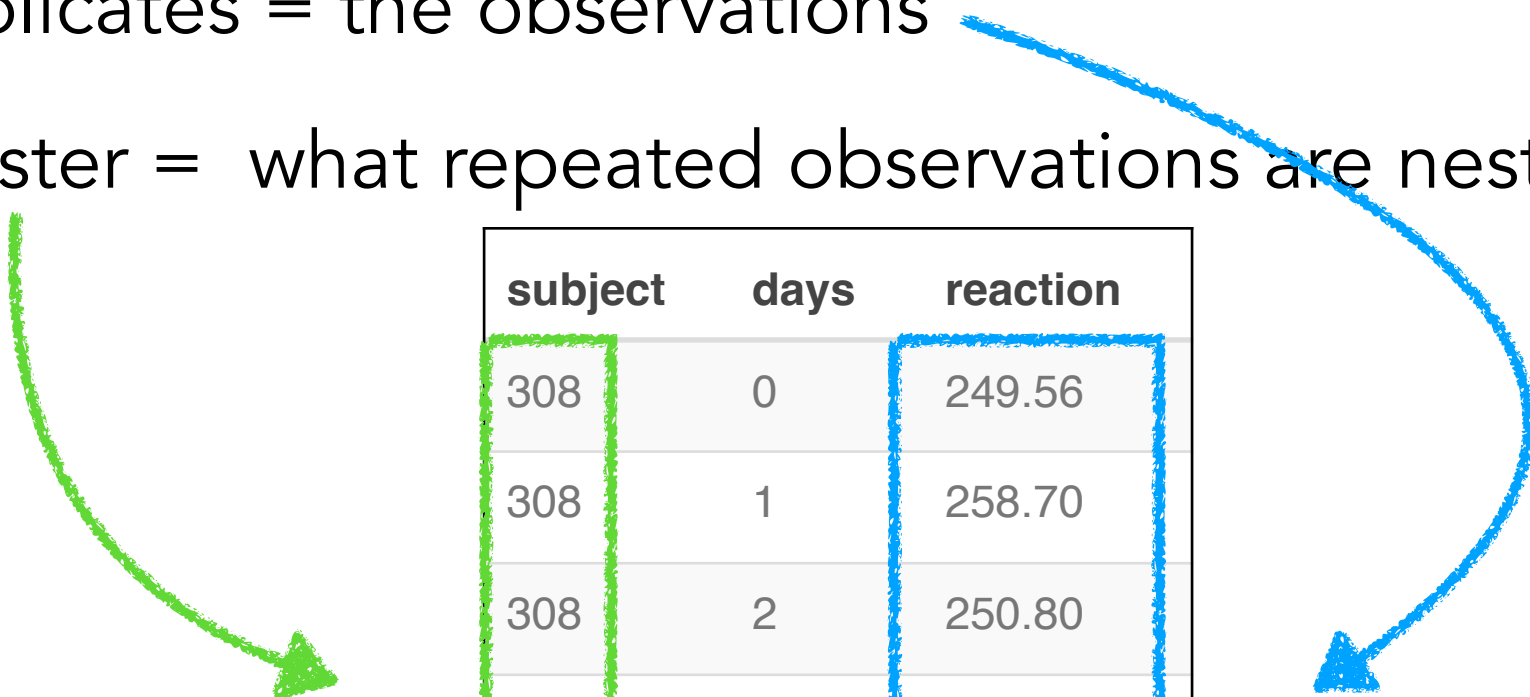
Varying intercepts and slopes



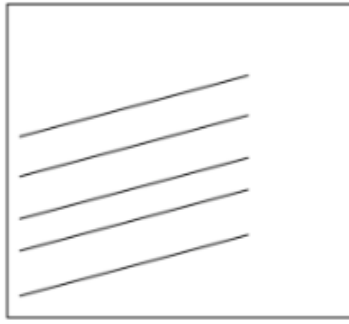
subject	days	reaction
308	0	249.56
308	1	258.70
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309	1	205.27
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309	3	204.71
309	4	207.72

Specifying Random Effects

- Random **Intercept**
 - Captures **baseline variation in DV** for each cluster of replicates
 - Replicates = the observations
 - Cluster = what repeated observations are nested inside of

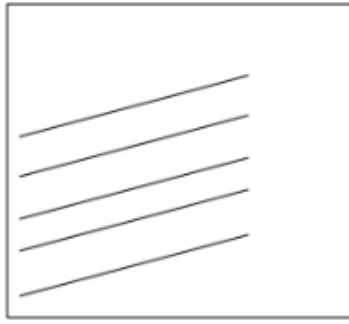


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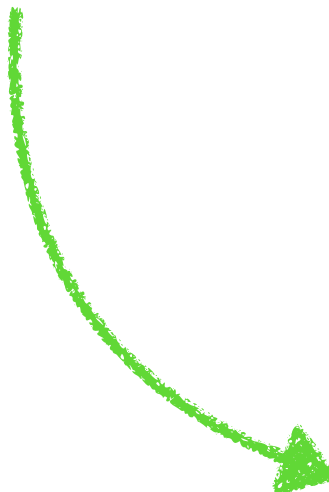


Specifying Random Effects

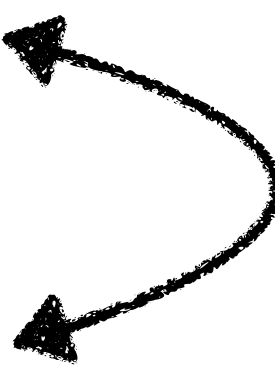
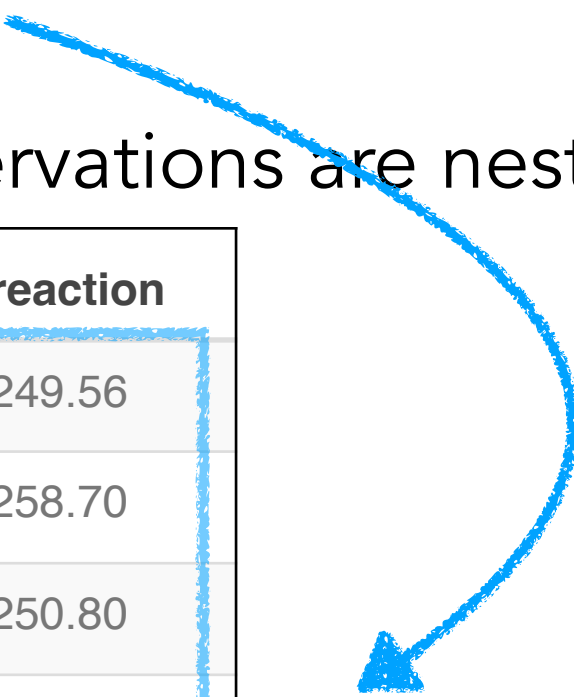
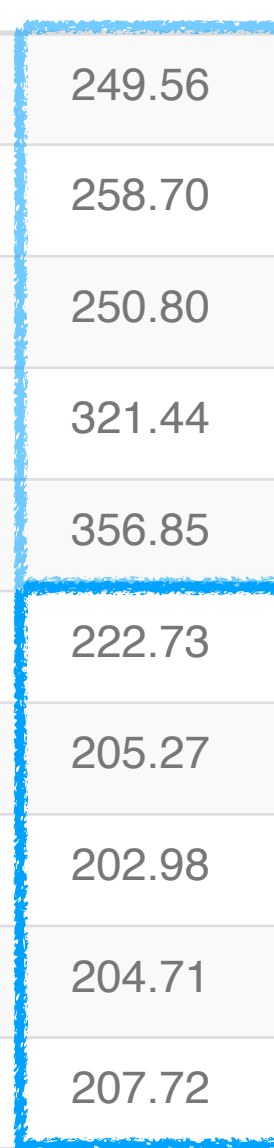
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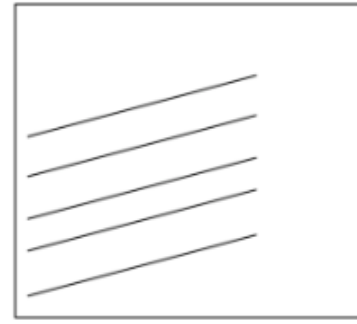
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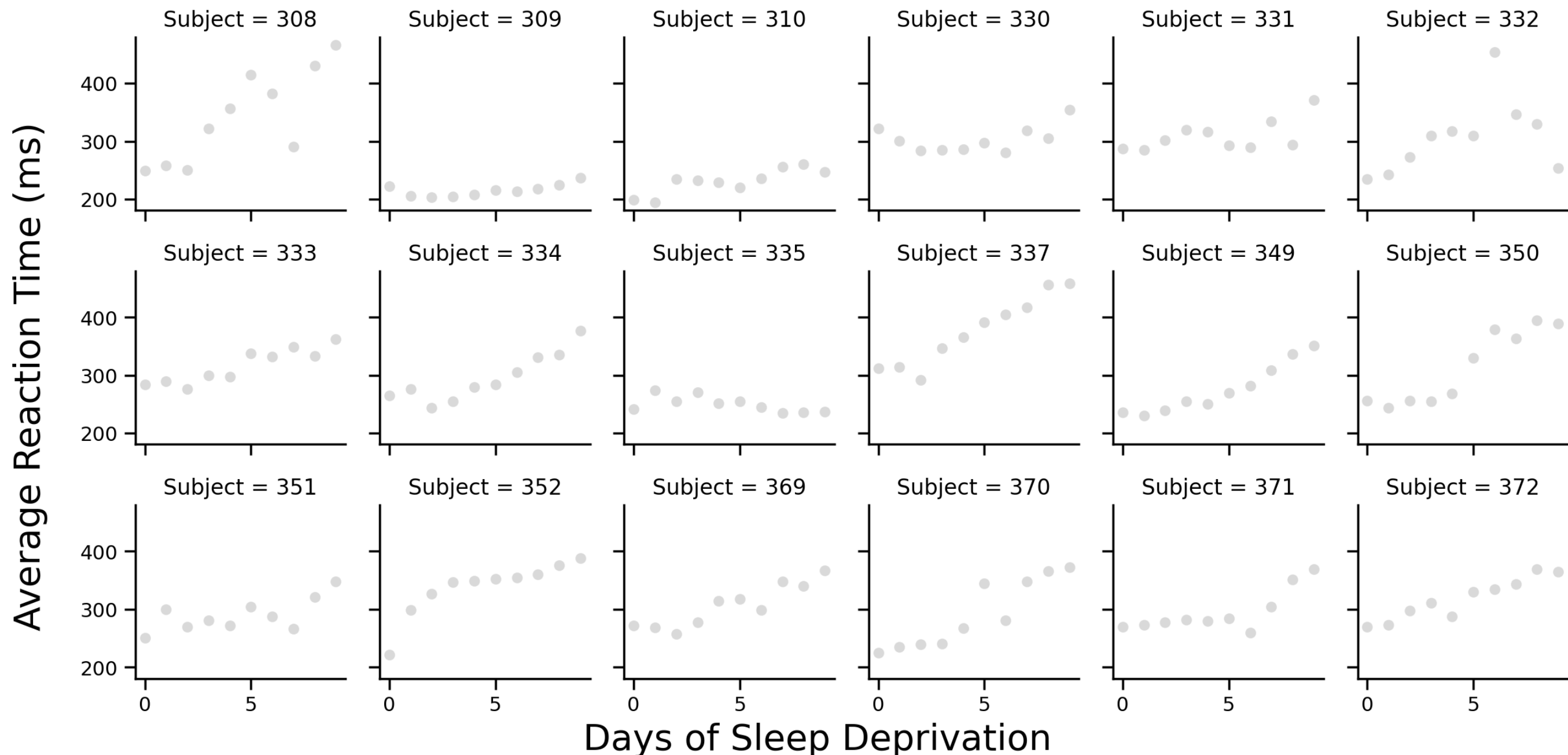
Account for the fact that these subjects have different average RTs

Specifying Random Effects

Varying intercepts

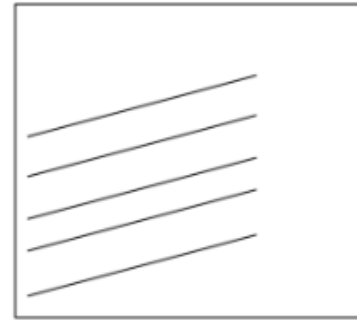


- Random **Intercept**
 - Captures **baseline variation in DV** for each cluster of replicates

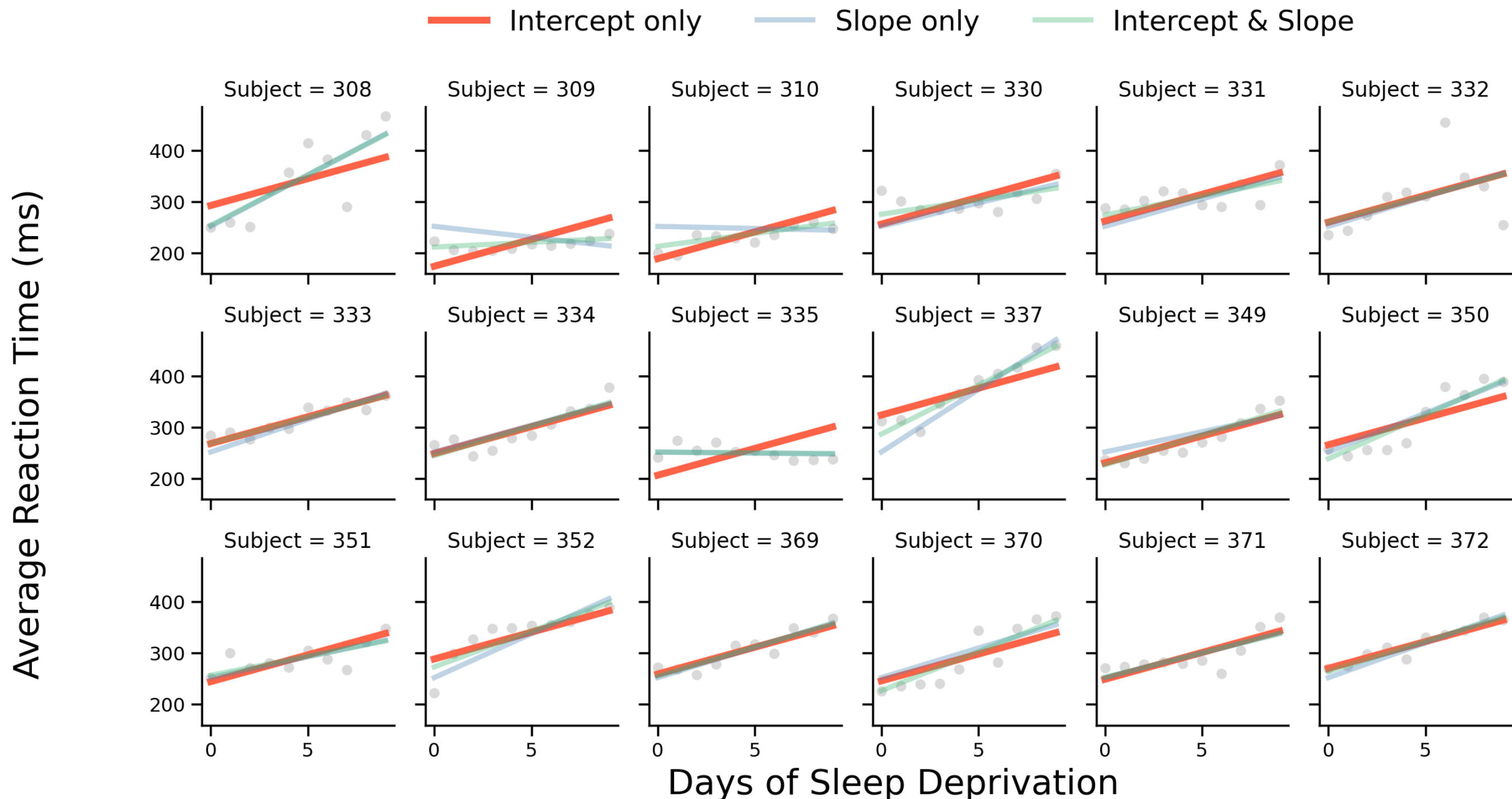


Specifying Random Effects

Varying intercepts

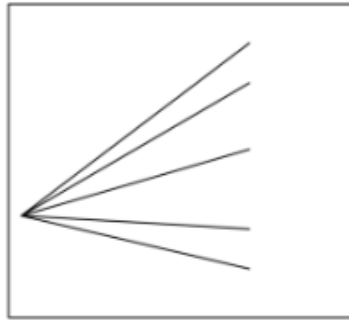


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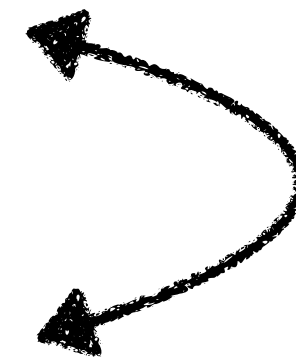
Specifying Random Effects

Varying slopes



- Random **Slope**
- Captures **variation in effect of IV(s)** for each cluster of replicates

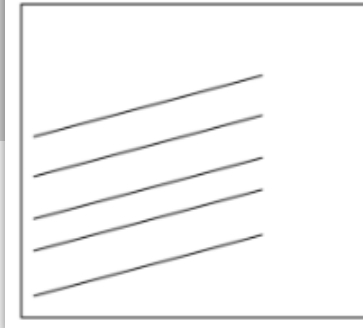
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Account for the fact
that these subjects
have different
relationships
between RT ~ day

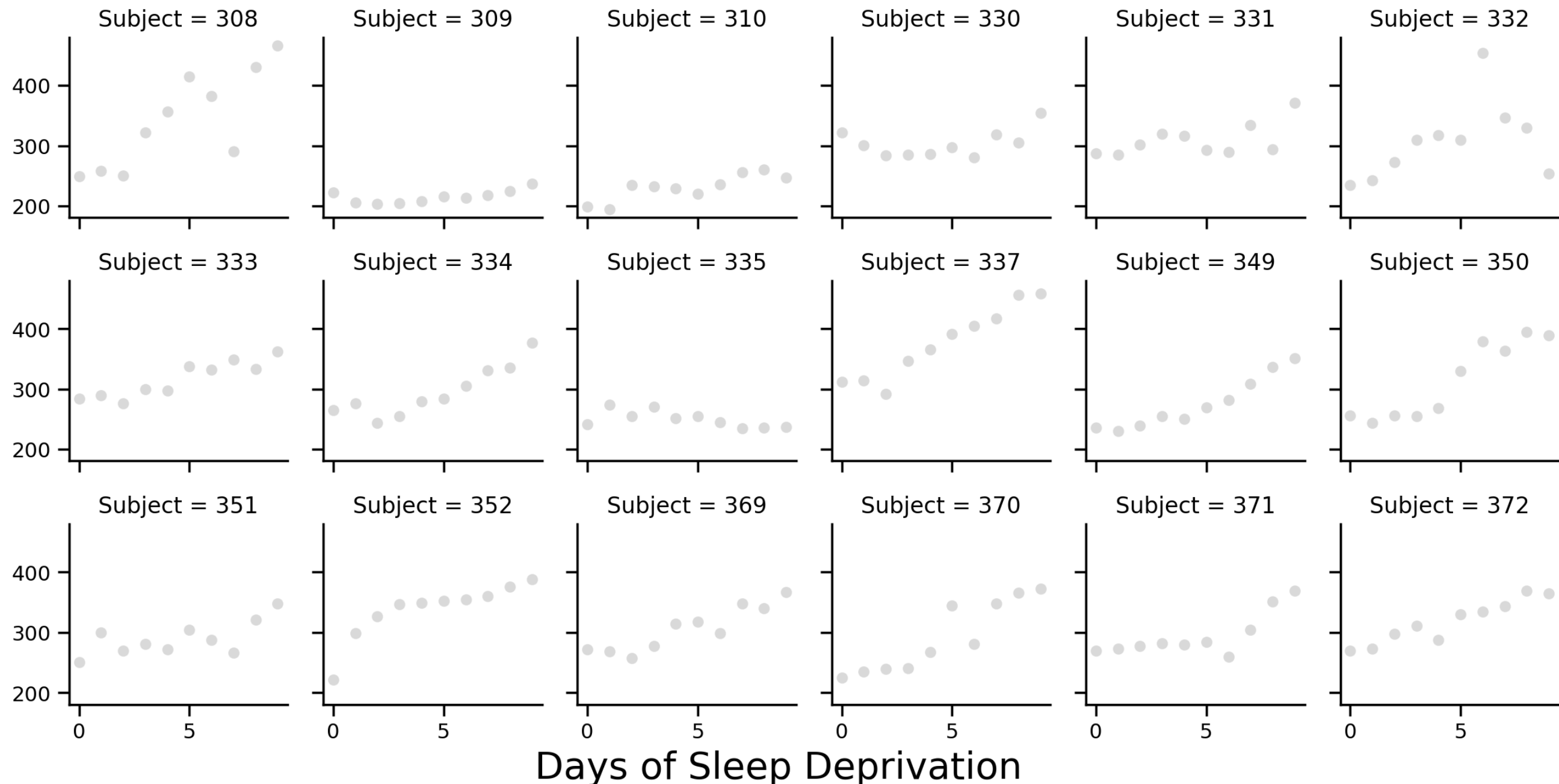
Specifying Random Effects

Varying intercepts



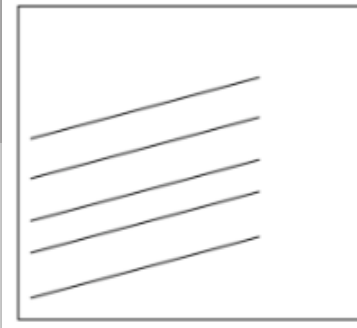
- Random **Slope**
 - Captures **variation in effect of IV(s)** for each cluster of replicates

Average Reaction Time (ms)

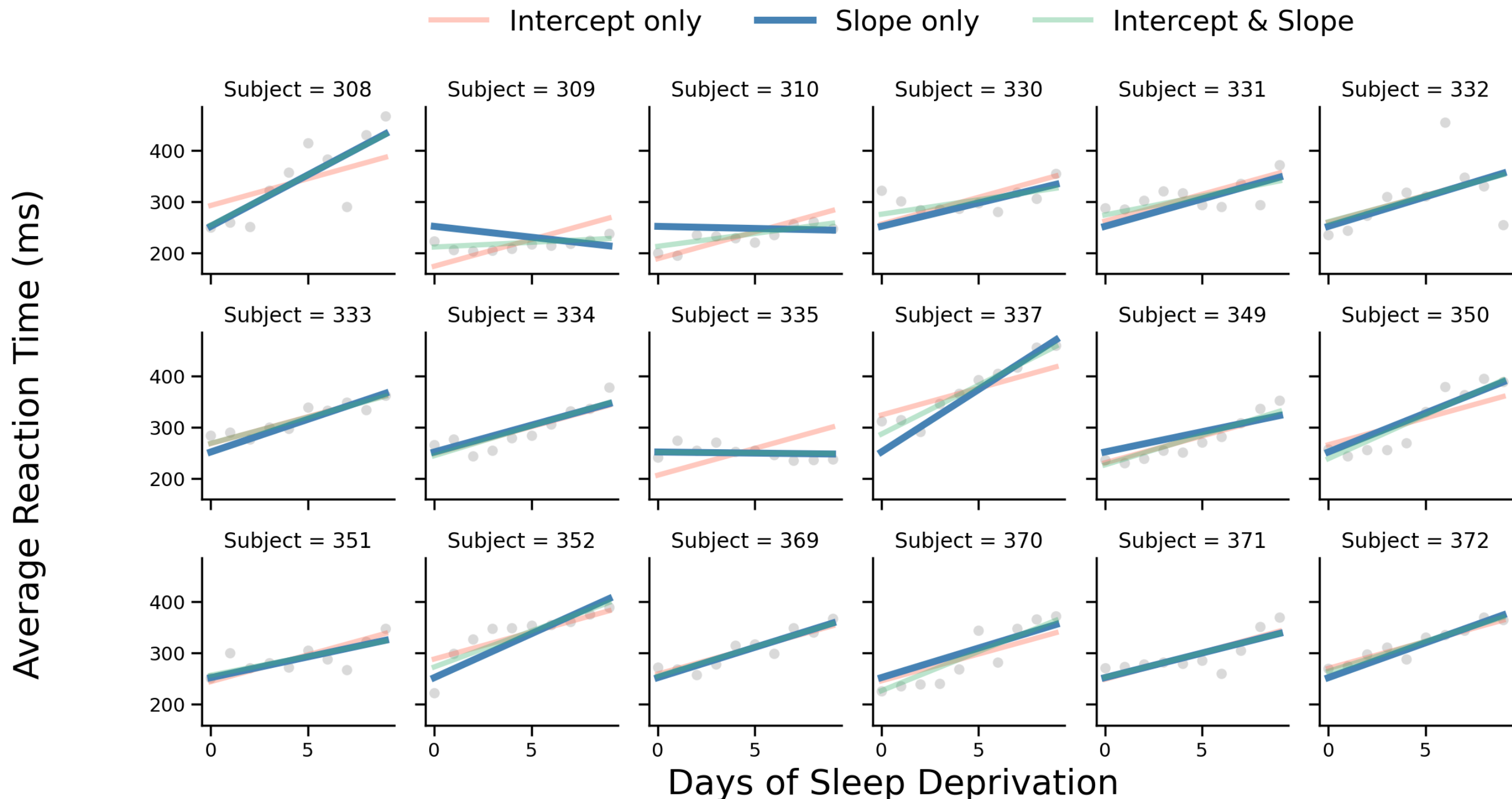


Specifying Random Effects

Varying intercepts

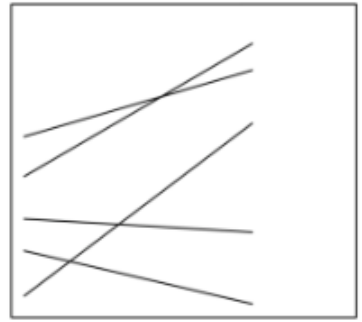


- Random **Slope**
- Captures **variation in effect of IV(s)** for each cluster of replicates



Specifying Random Effects

Varying intercepts and slopes

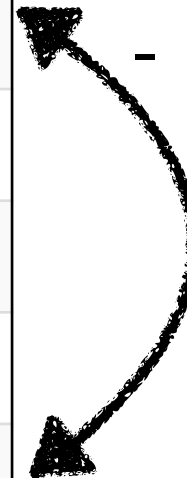


- Random **Intercept + Slope** (& typically correlation)
- Captures **variation in both baseline DV, effect of IV(s), and their correlation** for each cluster of replicates

subject	days	reaction
308	0	249.56
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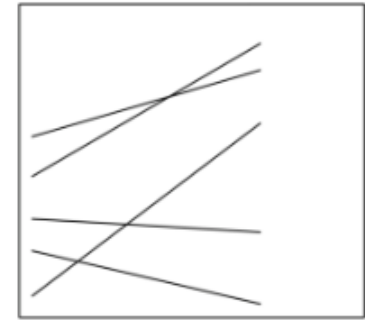
Account for:

- different baseline RTs
- different relationships between RT ~ day
- different correlation between baseline RT and RT ~ day
 - e.g. larger RTs = steeper slope
 - *slower* Subjects show *stronger* effect of Sleep Deprivation

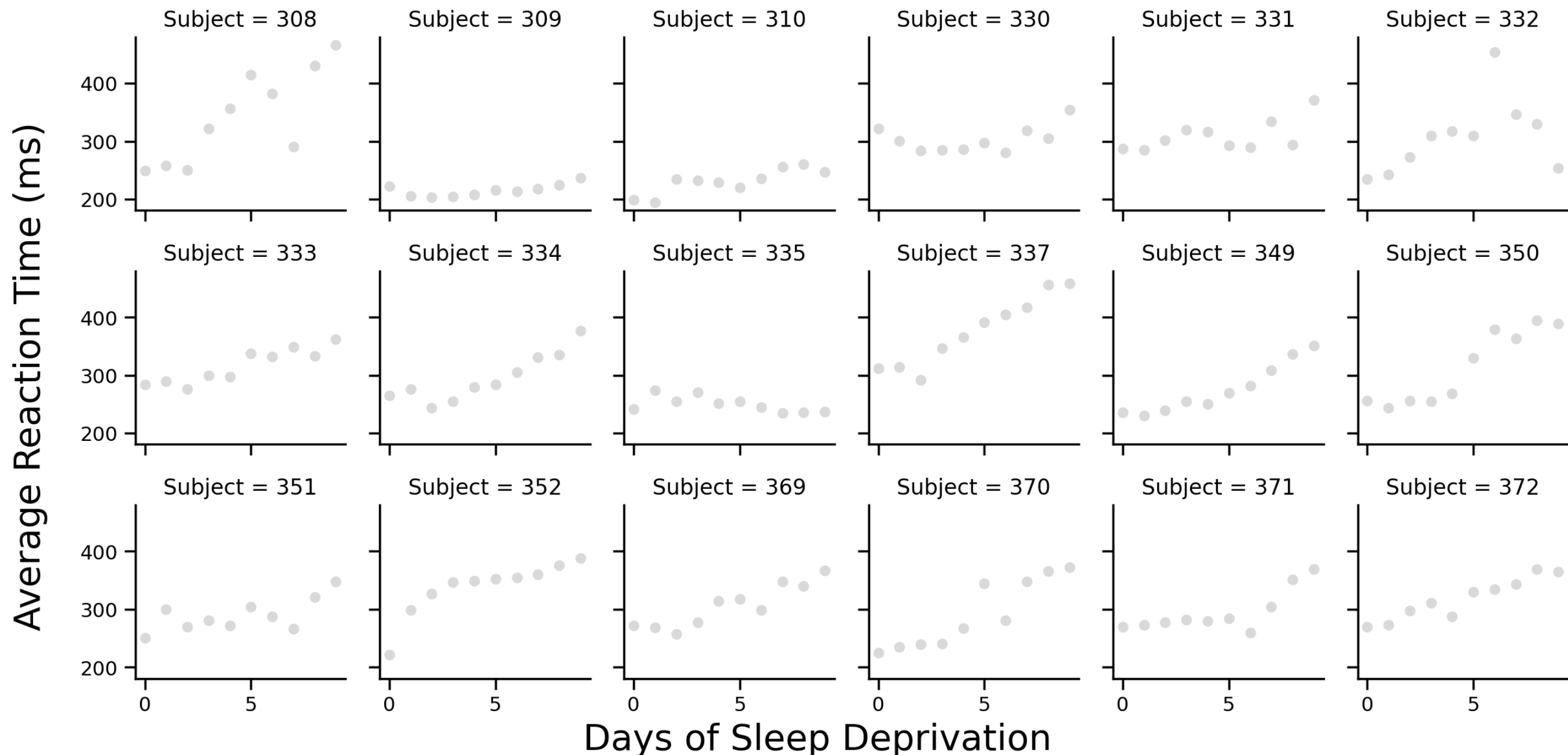


Specifying Random Effects

Varying intercepts and slopes

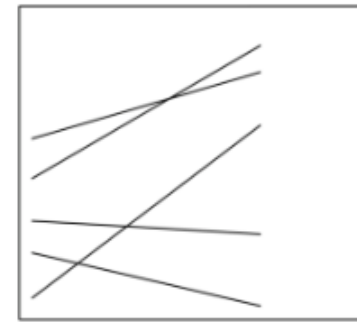


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Specifying Random Effects

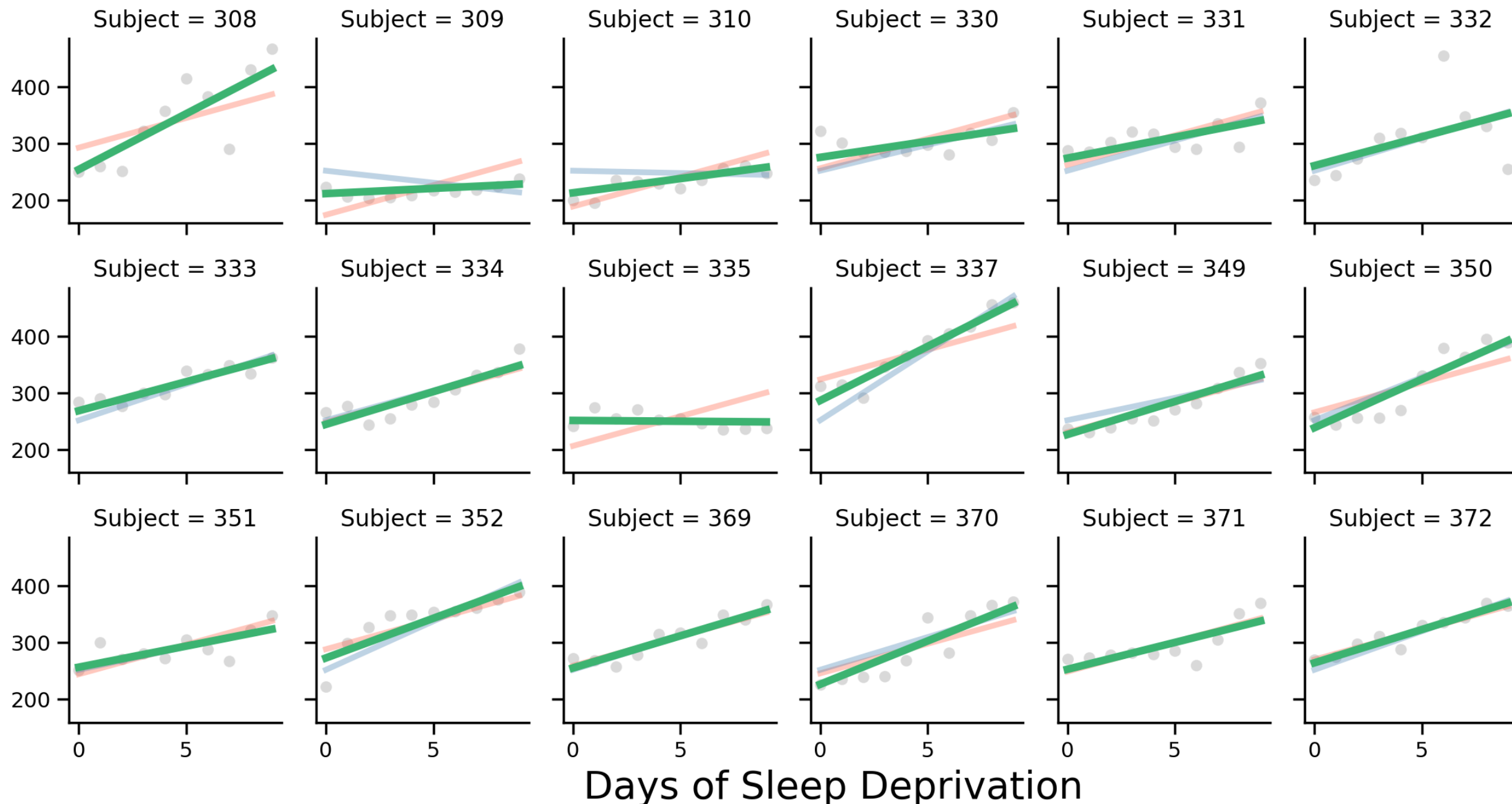
Varying intercepts and slopes



- Random **Intercept + Slope** (& typically correlation)
- Captures **variation in both baseline DV, effect of IV(s), and their correlation** for each cluster of replicates

— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



What should I include as Random Effects?

General Advice: **Start Maximal** then prune down



Contents lists available at [SciVerse ScienceDirect](#)

Journal of Memory and Language

journal homepage: www.elsevier.com/locate/jml

Random effects structure for confirmatory hypothesis testing:
Keep it maximal

Dale J. Barr^{a,*}, Roger Levy^b, Christoph Scheepers^a, Harry J. Tily^c

^aInstitute of Neuroscience and Psychology, University of Glasgow, 58 Hillhead St., Glasgow G12 8QB, United Kingdom

^bDepartment of Linguistics, University of California at San Diego, La Jolla, CA 92093-0108, USA

^cDepartment of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

*"Through theoretical arguments and Monte Carlo simulation, we show that LMMs generalize best when they include the **maximal random effects structure justified by the design...***

***Maximal LMMs should be the 'gold standard'** for confirmatory hypothesis testing in psycholinguistics and beyond."*

- Failure to include maximal random-effect structures in LMMs (when such random effects are present in the underlying populations) **inflates Type I error rates**
- For designs including within-subjects (or within-items) manipulations, random-**intercepts-only LMMs can have catastrophically high Type I error rates**, regardless of how p-values are computed from them
- In terms of power, maximal models perform surprisingly well even in a "worst case" scenario where they assume random slope variation that is actually not present in the population

What should I include as Random Effects?

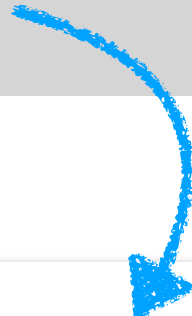
General Advice: **Start Maximal** then prune down

- Start with random intercepts, slopes, & correlations for each repeated factor you want to generalize over (e.g. Subject, Stimulus Category, etc)
 - **Reduce complexity** of the random effects structure **step-by-step**
 - Remove the correlation between random effects first, then slopes, last intercepts
 - **Examine the variance** across your clusters
 - Sometimes you don't have enough variance to estimate random effect; model will ***fail to converge***
 - Use **model comparison** to see whether extra random effects parameters are **worth it**
- Lab notebook(s) will cover these!

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 - Complete, No-pooling, **Partial-pooling**
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 - What's happening under-the-hood
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 - Random Intercepts & Slopes
 - How to choose your random-effects
4. How to estimate LMMs
 - Meet Pymer4

Estimating LMMs: Python vs R



Fitting Linear Mixed-Effects Models Using lme4

Douglas Bates Martin Mächler Benjamin M. Bolker Steven C. Walker
University of Wisconsin-Madison ETH Zurich McMaster University McMaster University

Abstract

Maximum likelihood or restricted maximum likelihood (REML) estimates of the parameters in linear mixed-effects models can be determined using the `lmer` function in the `lme4` package for R. As for most model-fitting functions in R, the model is described in

`lme4` is the **gold standard** for estimating LMMs

But we're using Python?!

Estimating LMMs: Python vs R

Fitting Linear Mixed-Effects Models Using lme4

Douglas Bates Martin Mächler Benjamin M. Bolker Steven C. Walker
University of Wisconsin-Madison ETH Zurich McMaster University McMaster University

Abstract

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Pymer4

Tests **passing** Build and Deploy **passing** pypi package
Platforms **noarch,osx-64,win-64,linux-64** downloads **197k**
python **3.8 | 3.9 | 3.10 | 3.11**
contributions **welcome**

Pymer4: Connecting R and Python for Linear Mixed Modeling

Eshin Jolly¹

¹ Dartmouth College



`pymer4` is a statistics library for estimating various regression and multi-level models in Python. Love `lme4` in R, but prefer to work in the scientific Python ecosystem? This package has got you covered!

`pymer4` provides a clean interface that hides the back-and-forth code required when moving between R and Python. In other words, you can work completely in Python, never having to deal with R, but get (most) of `lme4`'s goodness. This is accomplished using `rpy2` to interface between languages.

Additionally `pymer4` can fit various additional regression models with some bells, such as robust standard errors, and two-stage regression (summary statistics) models. See the features page for more information.

TL;DR This package is your new *simple* Pythonic drop-in replacement for `lm()` or `glmer()` in R.

For an example of what's possible check out the tutorials or [this blog post](#) comparing different modeling strategies for clustered/repeated-measures data.

Estimating LMMs: Python vs R

Brings lme4 functionality to Python (with some slight differences):

- Auto-includes p-value calculations that lme4 omits (via lmerTest)
- Auto-includes 95% CIs in summary; need to use `confint()` in R
- Easier to work with fixed/random effects, fits, and residuals

Pymer4

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Build and Deploy **passing**

pypi package

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Estimating LMMs: Python vs R

Brings lme4 functionality to Python (with some slight differences):

Python	R	Description
<code>Lmer()</code>	<code>lmer()</code>	Creating model
<code>model.fit()</code> and <code>model.summary()</code>	<code>summary(model)</code>	Estimating & Summary
<code>model.anova()</code>	<code>anova(model)</code>	ANOVA F-tests
<code>.fit(factors={'cond':{'A':1,;B':-1}})</code>	<code>contrasts()</code>	Categorical variables
<code>lrt(m1, m2)</code>	<code>anova(m1, m2)</code>	To compare models

But formula syntax is the same! `"Reaction ~ Days + (1 + Days | Subject)"`

Pymer4

Tests **passing**

Build and Deploy **passing**

pypi package

Platforms **noarch,osx-64,win-64,linux-64**

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Pymer4: Connecting R and Python for Linear Mixed Modeling

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Estimating LMMs: Syntax & Coefficients

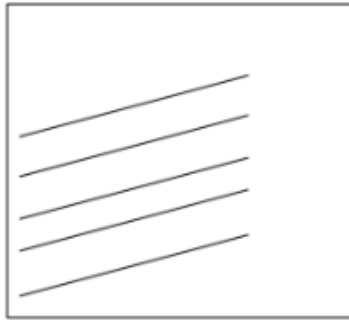
But formula syntax is the same! `Reaction ~ Days + (1 + Days | Subject)`

Random Effect Syntax Conventions

Formula (Random effects part only)	Description
<code>(1 Subject)</code>	Random intercept for each Subject
<code>(0 + X1 Subject)</code>	Random slope for each Subject
<code>(X1 Subject)</code> <code>(1 + X1 Subject)</code>	Correlated random intercept & slope for each Subject
<code>(X1 Subject)</code> <code>(1 Subject) + (0 + X1 Subject)</code>	Uncorrelated random intercept & slope for each Subject
<code>(X1 + X2 Subject)</code>	Correlated random intercept & slope for each Subject for multiple predictors
<code>(X1 * X2 Subject)</code>	Same but include interaction too
<code>(X1 Subject) + (X2 Stimulus)</code>	Correlated intercepts and slopes for multiple repeated variables

Estimating LMMs: Syntax & Coefficients

Varying intercepts



Random Intercept

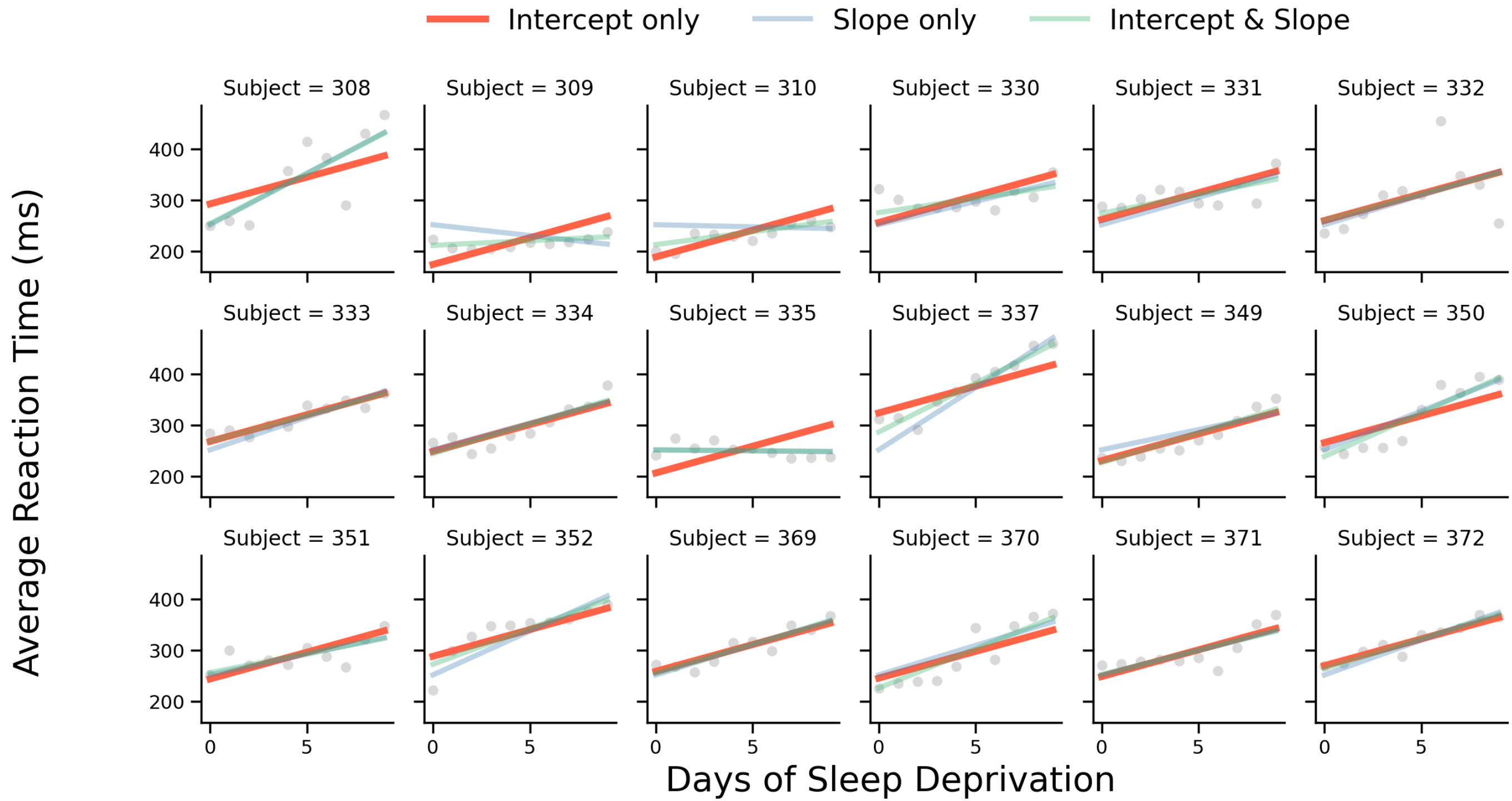
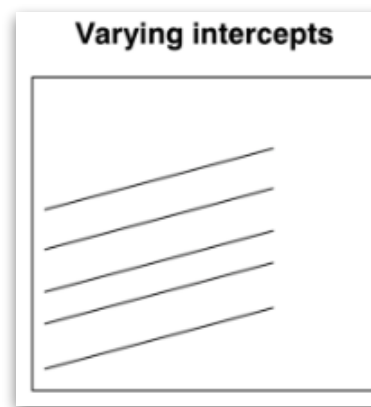
```
Lmer ("Reaction ~ Days + (1 | Subject)",  
data = df.to_pandas())
```

Formula (Random effects part only)	Description
(1 Subject)	Random intercept for each Subject
(0 + X1 Subject)	Random slope for each Subject
(X1 Subject) (1 + X1 Subject)	Correlated random intercept & slope for each Subject
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Estimating LMMs: Syntax & Coefficients

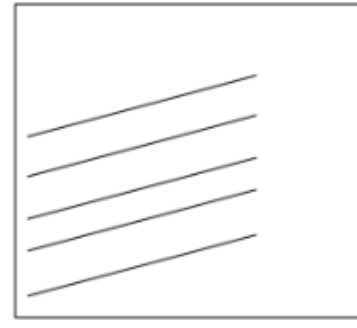
Random Intercept

```
lmer("Reaction ~ Days + (1 | Subject)",  
data = df.to_pandas())
```



Estimating LMMs: Syntax & Coefficients

Varying intercepts



Random Intercept

```
lmer("Reaction ~ Days + (1 | Subject)",  
data = df.to_pandas())
```

Fixed Effects Coefficients

	Estimate	2.5_ci	97.5_ci	SE	DF	T-stat	P-val	Sig
(Intercept)	251.405105	232.301892	270.508318	9.746716	22.810199	25.793826	2.241351e-18	***
Days	10.467286	8.891041	12.043531	0.804221	161.000000	13.015428	6.412601e-27	***

Remember the data

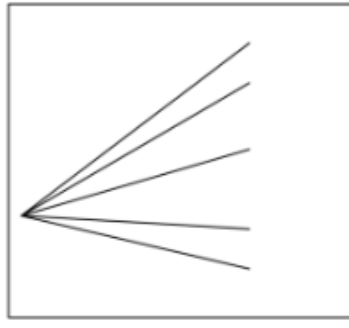
subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72

	(Intercept)	Days
308	292.188815	10.467286
309	173.555551	10.467286
310	188.296537	10.467286
330	255.811547	10.467286
331	261.621294	10.467286
332	259.626343	10.467286
333	267.905599	10.467286
334	248.408124	10.467286
335	206.122978	10.467286
337	323.587791	10.467286
349	230.208856	10.467286
350	265.516468	10.467286
351	243.542884	10.467286

Random Effects Coefficients
Different Intercept per Subject

Estimating LMMs: Syntax & Coefficients

Varying slopes



Random Slope

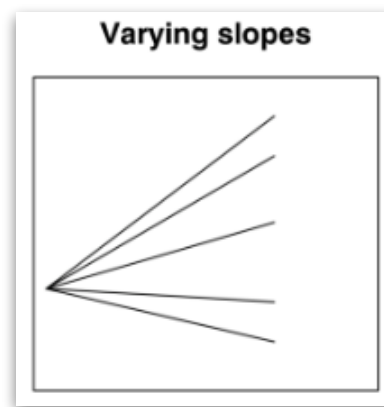
```
Lmer ("Reaction ~ Days + (0 + Days | Subject)",  
data = df.to_pandas())
```

Formula (Random effects part only)	Description
(1 Subject)	Random intercept for each Subject
(0 + X1 Subject)	Random slope for each Subject
(X1 Subject) (1 + X1 Subject)	Correlated random intercept & slope for each Subject
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Estimating LMMs: Syntax & Coefficients

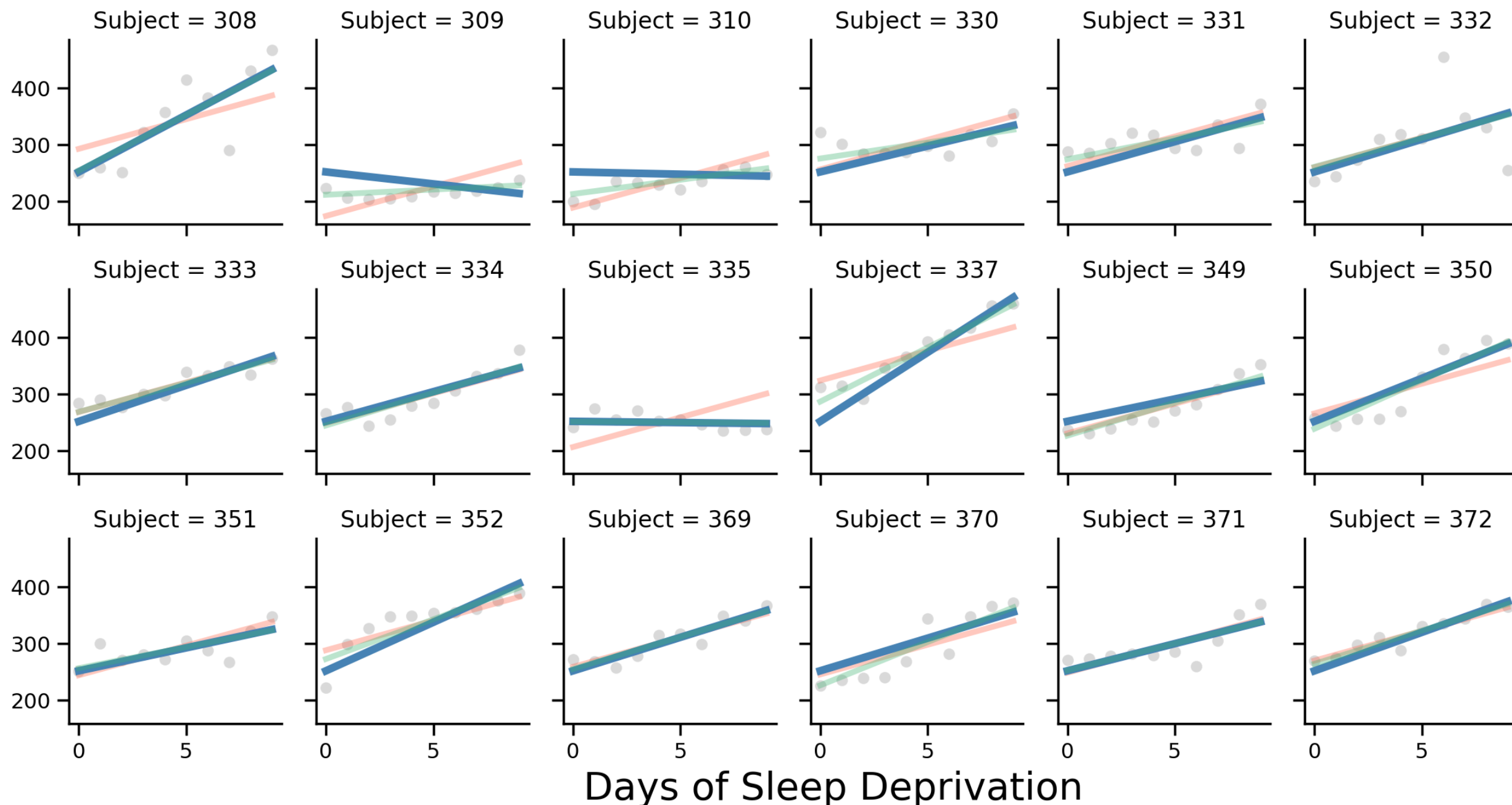
Random Slope

```
Lmer ("Reaction ~ Days + (0 + Days | Subject)",  
data = df.to_pandas())
```



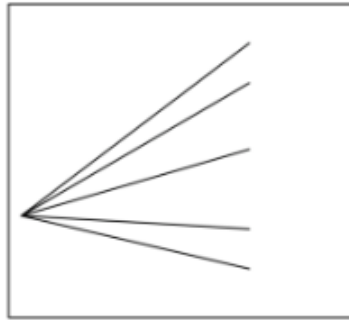
— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



Estimating LMMs: Syntax & Coefficients

Varying slopes



Random Slope

```
Lmer("Reaction ~ Days + (0 + Days | Subject)",  
data = df.to_pandas())
```

Fixed Effects Coefficients

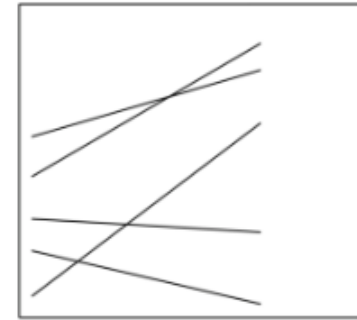
	Estimate	2.5_ci	97.5_ci	SE	DF	T-stat	P-val	Sig
(Intercept)	251.405105	243.526107	259.284103	4.019971	161.000003	62.539039	7.333827e-115	***
Days	10.467286	6.803019	14.131552	1.869558	21.679342	5.598802	1.319902e-05	***

	(Intercept)	Days
308	251.405105	20.086692
309	251.405105	-4.232671
310	251.405105	-0.818920
330	251.405105	9.127388
331	251.405105	10.675484
332	251.405105	11.535298
333	251.405105	12.743009
334	251.405105	10.477487
335	251.405105	-0.433739
337	251.405105	24.357732
349	251.405105	7.906925
350	251.405105	15.201214
351	251.405105	8.104156

Random Effects Coefficients
Different Slopes per Subject

Estimating LMMs: Syntax & Coefficients

Varying intercepts and slopes



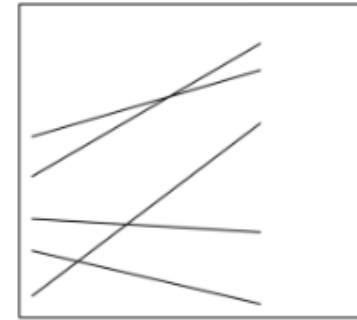
Random Intercept, Slope, & Correlation

```
Lmer ("Reaction ~ Days + (1 + Days | Subject)",  
data = df.to_pandas())
```

Formula (Random effects part only)	Description
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Estimating LMMs: Syntax & Coefficients

Varying intercepts and slopes



Random **Intercept**, **Slope**, & **Correlation**

OR:

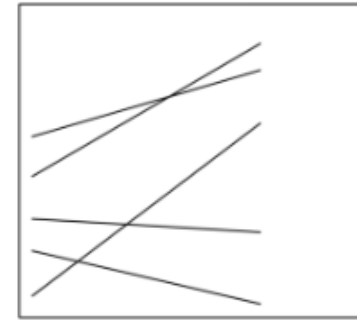
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Estimating LMMs: Syntax & Coefficients

Random Intercept, Slope, & Correlation

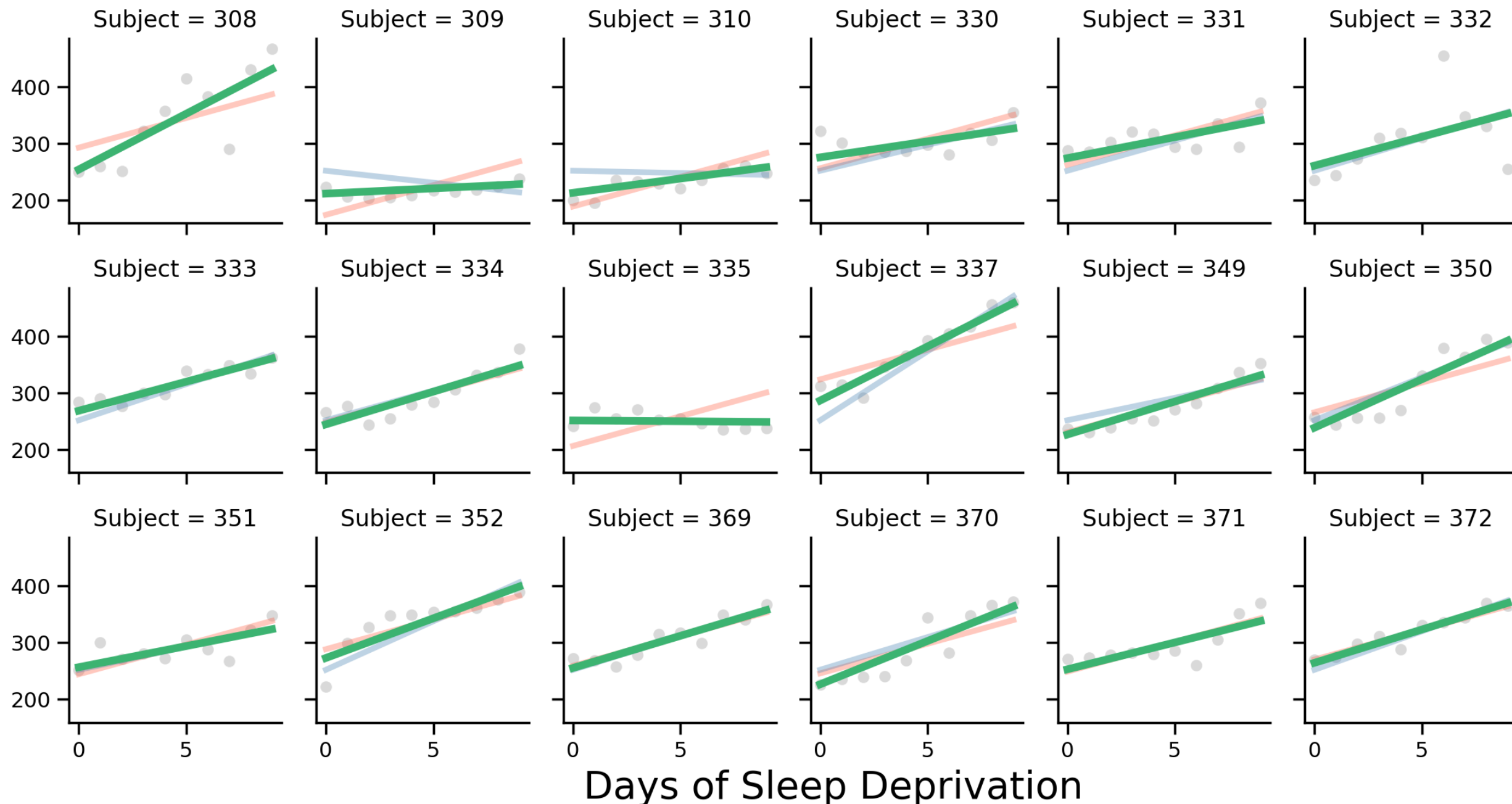
Varying intercepts and slopes



```
Lmer ("Reaction ~ Days + (Days | Subject)",  
data = df.to_pandas())
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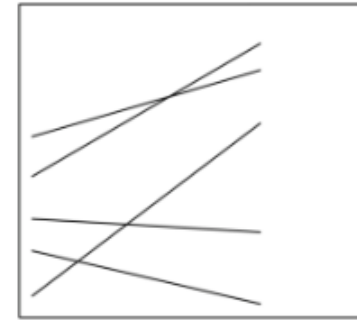
— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



Estimating LMMs: Syntax & Coefficients

Varying intercepts and slopes



Random Intercept, Slope, & Correlation

```
Lmer("Reaction ~ Days + (Days | Subject)",  
data = df.to_pandas())
```

Fixed Effects Coefficients

	Estimate	2.5_ci	97.5_ci	SE	DF	T-stat	P-val	Sig
(Intercept)	251.405105	238.029141	264.781069	6.824597	16.999726	36.838090	1.171558e-17	***
Days	10.467286	7.437594	13.496978	1.545790	16.999984	6.771481	3.263824e-06	***

	(Intercept)	Days
308	253.663656	19.666262
309	211.006367	1.847605
310	212.444696	5.018429
330	275.095724	5.652936
331	273.665417	7.397374
332	260.444673	10.195109
333	268.245613	10.243650
334	244.172490	11.541868
335	251.071436	-0.284879
337	286.295592	19.095551
349	226.194876	11.640718
350	238.335071	17.081504
351	255.982969	7.452024

Random Effects Coefficients
Different Intercept **AND** Slopes
by Subject

Estimating LMMs: Syntax & Coefficients

Syntax conventions for specifying random effects

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Estimating LMMs: Syntax & Coefficients

You can also specify random effects for multiple variables

We'll explore more in notebooks

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LMMs: The Essentials

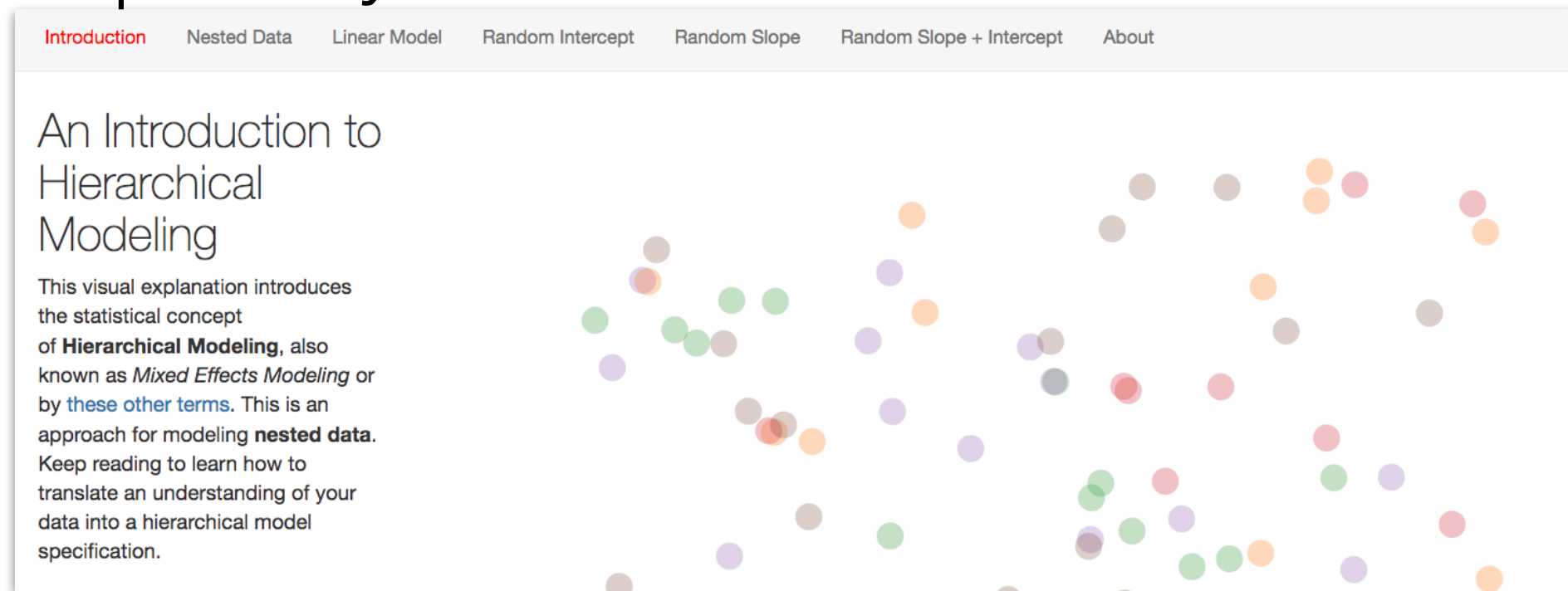
1. Ways to aggregate dependent data
 - Complete, No-pooling, Partial-pooling
2. Parts of the mixed model
 - Fixed vs Random Effects
 - What's happening under-the-hood
3. Specifying Random Effects
 - Random Intercepts & Slopes
 - How to choose your random-effects
4. How to estimate LMMs
 - Meet Pymer4

LMMs: The Essentials

1. Ways to aggregate dependent data
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 - Random **Intercepts** & **Slopes**
 - How to choose your random-effects
4. How to estimate LMMs
 - Meet **Pymer4**

Let's take a 15m Break, then:

1. Checkout this short interactive visualization
<https://tinyurl.com/201b-lmm>



2. Clone Github Assignment for this week and work through 01_lmms